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Adult neurogenesis promotes pattern separation and sparsity beyond the dentate gyrus in a spiking DG–CA3 model

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List of abbreviations and acronyms

BC	Basket Cell
CA1	Cornu Ammonis area 1
CA3	Cornu Ammonis area 3
DG	Dentate Gyrus
EC	Entorhinal Cortex
GC	Granule Cell
HIPP	Hilar Perforant Path-associated cell
iCA3	inhibitory CA3 cell
iGC	immature Granule Cell
MC	Mossy Cell
mGC	mature Granule Cell
pCA3	pyramidal CA3 cell
PP	Perforant Path
PSTH	Peri-Stimulus Time Histogram
SEM	Standard Error of the Mean
STDP	Spike-Timing-Dependent Plasticity

Resumo

O giro denteado (DG) é responsável pela separação de padrões, função computacional que transforma padrões de entrada similares em representações neurais mais distintas, e é também uma das únicas áreas do encéfalo de mamíferos que gera novos neurônios ao longo de toda a vida adulta. As células granulares (GCs) geradas por essa neurogênese passam por uma fase imatura de alta excitabilidade, e ativá-las *in vivo* recruta interneurônios inibitórios e aumenta a esparsidade da atividade tanto no DG quanto no Cornu Ammonis 3 (CA3). Ainda não foi investigado, no entanto, se esse efeito sobre o circuito decorre do microcircuito já conhecido do DG-CA3, visto que os modelos computacionais existentes de neurogênese e separação de padrões se restringem ao DG. Neste trabalho, construiu-se um modelo computacional da rede DG-CA3 a partir da base de conhecimento Hippocampome.org, estendido com uma população de GCs imaturas (iGCs). A separação de padrões e a esparsidade foram medidas variando-se o nível de neurogênese e a excitabilidade das iGCs, que em seguida foram ativadas diretamente. A neurogênese aumentou a capacidade de separação de padrões das GCs maduras (mGCs) e levou o CA3 no mesmo sentido, principalmente por tornar a atividade populacional mais esparsa; já a ativação direta das iGCs recrutou interneurônios inibitórios por todo o DG e o CA3, aumentando a esparsidade e reproduzindo o efeito observado *in vivo*. Conclui-se que a influência da neurogênese adulta sobre a esparsidade pode emergir apenas desse microcircuito e que ela alcança regiões além do próprio DG, corroborando a visão das iGCs como moduladoras da atividade do circuito, ao invés de codificadoras de informação.

Palavras-chave: Neurogênese adulta, separação de padrões, giro denteado, CA3, rede neural pulsada, esparsidade.

Abstract

The dentate gyrus (DG) transforms similar inputs into more distinct representations, a computation termed pattern separation, and is one of the few brain regions that generate new neurons throughout adulthood. These adult-born granule cells (GCs) pass through an immature, highly excitable stage, and activating them *in vivo* recruits inhibitory interneurons and increases population sparsity across both the DG and CA3 (Cornu Ammonis area 3). Whether this circuit-wide effect follows from the known DG-CA3 microcircuitry has not been examined in a biophysically detailed model, and existing models of neurogenesis and pattern separation are confined to the DG. Here, a spiking model of the DG-CA3 network was built from the Hippocampome.org knowledge base and extended with a population of immature GCs (iGCs). Pattern separation and population sparsity were measured while the level of neurogenesis and iGC excitability was varied, and the iGCs were then activated directly. Neurogenesis enhanced pattern separation in the mature GCs (mGCs) and pushed CA3 in the same direction, mainly by making population activity sparser; direct activation of the iGCs recruited inhibitory interneurons throughout the DG and CA3 and increased population sparsity, reproducing the effect observed *in vivo*. The sparsifying influence of adult neurogenesis can therefore emerge from the known DG-CA3 circuitry and reaches beyond the DG, consistent with a modulatory rather than an encoding role for iGCs.

Keywords: Adult neurogenesis, pattern separation, dentate gyrus, CA3, spiking neural network, sparsity.

1 Introduction

To the hippocampus is attributed a fundamental role in the encoding, consolidation, and recall of declarative memories (Eichenbaum, 1999; Liao & Losonczy, 2024). To this end, the hippocampus receives afferent inputs from multiple cortical association areas, where information from diverse sensory modalities is processed. This information converges through the parahippocampal region, which comprises the perirhinal, postrhinal, and, most importantly, the entorhinal cortex (EC) (Eichenbaum, 2000). This rich convergence of information positions the hippocampus ideally to encode associations between events, consolidating its central role in learning and memory (Berdugo-Vega et al., 2023; Henke, 2010).

In a simplified and traditional account, the structure of the hippocampus has been described as consisting primarily of the trisynaptic circuit: information flows in a single direction from the EC to the dentate gyrus (DG), to CA3, and to CA1. The EC serves as the main input to the DG via the perforant path (PP), whose neurons connect primarily with Granule Cells (GCs), the principal excitatory cells of the DG, in a ratio of $1 : 10$; it is worth noting that the EC also sends afferent inputs to CA3 and CA1, which challenges the simplified notion of the trisynaptic circuit (Basu & Siegelbaum, 2015). Precisely because of this divergent EC-to-DG connectivity, it is believed that the DG is responsible for a computational function called pattern separation, through its sparse activity and the spike rate of GCs (Hainmueller & Bartos, 2020; Kesner & Hopkins, 2006; Yassa & Stark, 2011). Pattern separation disambiguates similar information, enabling the identification and differentiation of distinct yet highly similar experiences, which is fundamental for avoiding catastrophic interference between memories stored in CA3 (Rolls, 2013). Of the large GC population in the granular layer of the DG, one of the densest regions of the brain in cell bodies (Amaral et al., 2007), only a small fraction of cells fires in response to any given input arriving from the EC (Chavlis et al., 2017; Kesner & Hopkins, 2006).

To illustrate this computation, consider the following oversimplified example: experiences from the same place, with the same people and objects present and following a similar routine, are likely encoded in the EC in a very similar manner; however, owing to pattern separation performed by the DG, small details such as a specific conversation or a news item on television

may be sufficient to elicit a neural activity pattern in the DG that is very distinct from those of past experiences, allowing the encoding of a new memory (Eichenbaum, 2004).

In contrast to the divergent projection from the EC to the DG, the projection from the DG to CA3 is highly convergent. The CA3 region is characterized by a network of recurrent connections, where its pyramidal CA3 cells (pCA3) excite both interneurons and other pCA3 cells. This architecture forms an autoassociative network that allows the storage of patterns as neural assemblies: groups of neurons that represent the same pattern by spiking together, formed through Spike-Timing-Dependent Plasticity (STDP), which potentiates the connections of neurons that spike together (Kopsick et al., 2024). Once an assembly is formed through the strengthening of synapses between its cells, it can subsequently be reactivated by a partial input signal from the original pattern, since activating a subpopulation of assembly cells will cause the remaining cells to become active due to the associations formed by synaptic potentiation. This process is known as pattern completion, and recent computational models demonstrate that this architecture allows robust memory retrieval even from degraded cues (Kopsick et al., 2024; Le Duigou et al., 2014). Via the Schaffer collaterals, CA3 projects to CA1, which acts as an interface between the hippocampus and the neocortex, where memories are consolidated and recalled (Bartsch et al., 2011).

Beyond its fundamental role in pattern separation, the DG is one of the only areas of the mammalian brain to exhibit adult neurogenesis, the formation of new neurons from stem cells following development, continuously generating Granule Cells throughout life (Boldrini et al., 2018; Dumitru et al., 2025). Adult neurogenesis in mammals occurs only in a few other specific areas, such as the olfactory bulb in rodents and possibly the striatum, amygdala, and hypothalamus in humans (Alonso et al., 2024; Jurkowski et al., 2020).

The GCs generated by adult neurogenesis take several weeks to fully develop in rodents. At 3 weeks, the vast majority of immature Granule Cells (iGCs) die, and from approximately 8 weeks onward they become practically indistinguishable from mature Granule Cells (mGCs) (Denoth-Lippuner & Jessberger, 2021). Over the course of maturation, iGCs undergo various morphological and electrophysiological changes, expanding their dendrites and forming connections with CA3. A critical period occurs at approximately 4–7 weeks, during

which iGCs, already partially integrated into the circuit, exhibit greater excitability and synaptic plasticity compared to mGCs (Aimone et al., 2014; Denoth-Lippuner & Jessberger, 2021; Zhao et al., 2006), offering a critical window for plasticity and memory encoding (Berdugo-Vega et al., 2023). Enhancing neurogenesis has been shown to improve behavioral pattern separation (Sahay et al., 2011), showing that these cells are central to the DG's discriminative function.

In general, adult neurogenesis is implicated in all phases of memory, including encoding, consolidation, recall, and even forgetting (Chavan et al., 2025). iGCs play a crucial role in novelty detection and pattern separation, enabling the discrimination between similar memories. Furthermore, recent evidence suggests a specialized role for these neurons in memory consolidation during REM sleep (Chavan et al., 2025) and in the consolidation process that generalizes memories, being essential for structural changes in CA3 circuits through the non-selective activation of neural assemblies via their high excitability (Ko et al., 2025).

Despite numerous studies demonstrating the role of adult neurogenesis in different hippocampal functions (Berdugo-Vega et al., 2023; Chavan et al., 2025), the computational mechanisms at the circuit level through which iGCs perform these functions remain poorly understood.

This central question unfolds into two main theoretical lines regarding the function of iGCs (Berdugo-Vega et al., 2023). The first hypothesis is that iGCs participate directly in memory encoding, integrating directly into long-term memory engrams and fulfilling a role similar to mGCs, but with their own peculiarities (Piatti et al., 2013). In this scenario, they would act in the encoding of new information, in pattern integration, or in the prevention of catastrophic interference with memories already established by mGCs. The second hypothesis, which has been explored more recently, proposes that iGCs function as temporary circuit-modulating elements. In this role, they could both regulate DG activity, increasing the sparsity of activation by inhibiting mGCs, and modulate CA3 activity (Aimone, 2016; Berdugo-Vega et al., 2023; Fölsz et al., 2023), rather than being directly recruited in memory encoding as mGCs are.

Given the difficulty of experimentation in the DG and the complexity of information processed in the hippocampus, computational models have been invaluable for understanding the role of neurogenesis (Aimone, 2016). Computational models vary in scale, biological

plausibility, the circuits modeled, and various other factors; therefore, a range of different results exists regarding the function and mechanisms of neurogenesis (Aimone, 2016; Berdugo-Vega et al., 2023). Existing models of neurogenesis and pattern separation, however, are confined entirely to the DG (Chavlis et al., 2017; Kim & Lim, 2024b; Yang et al., 2025), and so cannot investigate the influence of iGCs into CA3 and CA1.

This work investigates whether the circuit-wide sparsification observed *in vivo* (McHugh et al., 2022) emerges from the known DG-CA3 microcircuitry, using a biophysically detailed spiking model of the DG-CA3 network that includes a population of iGCs. Pattern separation and population sparsity were measured while the level of adult neurogenesis was varied, and the iGCs were then activated directly to probe their effect on the rest of the circuit. Neurogenesis enhanced pattern separation in the mGCs and pushed CA3 in the same direction, mainly by making the population activity sparser. Directly activating the iGCs recruited inhibitory interneurons throughout the DG and CA3 and increased population sparsity. Together, these results support the view of iGCs as modulators that regulate activity across the hippocampal network rather than as encoders confined to the DG.

2 Literature review

2.1 The dentate gyrus and pattern separation

Two properties are usually thought to underlie pattern separation in the DG: the expansion of a relatively small entorhinal population onto a much larger number of GCs, and the competitive network dynamics that keep only a few of those GCs active at any one time (Hainmueller & Bartos, 2020). The sparseness of GC activity has been confirmed by recordings from identified cells: fewer than 5% of a rodent's GCs are active while it explores a given environment, and those that are active typically fire in a single location (Diamantaki et al., 2016; Hainmueller & Bartos, 2020).

This sparseness is not a property of the GCs alone but of the competitive dynamics of the network, in which active GC ensembles suppress competing ensembles by recruiting disynaptic inhibition (Hainmueller & Bartos, 2020). The DG contains several GABAergic interneuron types that implement this inhibition (Degro et al., 2022). The best characterized are the parvalbumin-expressing cells, which include the basket cells and inhibit the perisomatic domain of the GCs, and the somatostatin-expressing cells, which target the distal GC dendrites near their PP input synapses (Hainmueller & Bartos, 2020). The functional consequence of losing this balance is clear: when a small number of GCs become abnormally dominant, the discriminative function of the region degrades (Scharfman & Myers, 2016), and the spatial reach of the inhibition determines how locally the network synchronizes (Strüber et al., 2017).

Two further features of the DG architecture matter for how this inhibition is organized. The first is the lamellar arrangement of the region, in which the GC population and its projections are organized into transverse slices that operate with partial independence (Sloviter & Lømo, 2012), although the interlamellar connectome is denser than the strict form of the hypothesis assumed (Pak et al., 2022). The second is the mossy cell population of the hilus, which receives excitation from the GCs and projects back to GCs and to hilar interneurons, so that its net effect on the GC population is inhibitory (Scharfman & Myers, 2013). Mossy cells contribute to pattern separation in their own right (GoodSmith et al., 2019), and modelling work attributes that

contribution specifically to the indirect feedback inhibition they recruit (Galloni et al., 2022).

It should be noted that the strongest form of the pattern separation hypothesis is not fully supported by recent recordings. In vivo studies of identified DG neurons mostly report gradual changes in firing rates between similar environments rather than the complete, non-overlapping remapping that models of full orthogonalization assume, and a substantial fraction of mature GCs appears to encode features common to several contexts (Hainmueller & Bartos, 2020). Reviews of DG function accordingly treat pattern separation as one of several operations the region supports, alongside novelty detection and the binding of information to spatial context (Kesner, 2018).

2.2 CA3: autoassociation and pattern completion

The recurrent collaterals of CA3 give the region attractor dynamics: an input resembling a stored pattern drives the network towards the stored state, which is the mechanism usually invoked to explain pattern completion (Migliore et al., 2019; Rolls, 2013, 2018). The same dynamics can also push CA3 into a different attractor when its inputs are sufficiently distinct from one another, so the output of the region is not simply a smoothed version of its input (Hainmueller & Bartos, 2020).

The relative weight of the two inputs to CA3 is worth noting, because it sets a limit on how much influence the DG can exert. An individual pCA3 cell receives inputs from thousands of EC and other pCA3 cells but from only some fifty GCs, and many of those GCs are silent during exploration (Hainmueller & Bartos, 2020). The influence of the GCs is therefore not simply a matter of excitatory drive: GC burst activity reliably recruits pCA3 cells, whereas low-frequency GC firing preferentially engages feedforward interneurons and so exerts a net inhibitory effect on CA3 (Hainmueller & Bartos, 2020). This asymmetry is the mechanism by which a small GC population can shape CA3 activity without dominating it.

CA3 is not the end of the circuit as far as the DG is concerned. A portion of the pCA3 population projects back to the hilus, a backprojection that contributes to pattern separation in the DG itself (Myers & Scharfman, 2011). Recent anatomical work continues to refine the picture: connectivity between pCA3 cells is organized by cell subtype rather than at random (Sammons

et al., 2025).

2.3 Adult hippocampal neurogenesis

That new neurons are added to the adult hippocampus was first reported from autoradiographic evidence in the rat (Altman & Das, 1965), although the finding met resistance for decades before becoming established in rodents (L. C. Abbott & Nigussie, 2020; Ming & Song, 2011). At least two niches are well regarded to sustain this process in the rodent brain: the subventricular zone, whose progeny migrate to the olfactory bulb and differentiate into inhibitory neurons, and the subgranular zone of the DG, which is the only one that generates excitatory glutamatergic cells and therefore the only one that adds principal neurons to a cortical circuit (Fares et al., 2019; Fölsz et al., 2023).

The rate of addition is low. Of all DG GCs, on the order of 0.2% are generated daily in rats and 0.06% in mice (Cameron & Mckay, 2001; Fölsz et al., 2023), and roughly half of the cells produced are eliminated by waves of programmed cell death during maturation (Fölsz et al., 2023). Because this addition is balanced by the continuous removal of mostly perinatally generated mGCs, the total number of GCs stays approximately constant (Fölsz et al., 2023). The cells that do survive are stably maintained and eventually become indistinguishable from developmentally born GCs (Denoth-Lippuner & Jessberger, 2021; Fölsz et al., 2023).

Maturing cells extend their dendrites outwards through the molecular layer and send an axon towards CA3 (Zhao et al., 2006), and their electrophysiological properties change in step, recapitulating aspects of embryonic development (Espósito et al., 2005). During the critical window described in Chapter 1 they are more excitable than mGCs and also more plastic, showing a lower threshold for the induction of long-term potentiation and a larger potentiation amplitude (Aimone et al., 2014; Fölsz et al., 2023). The boundaries of this window, however, are not settled: birthdating studies combined with immediate-early gene expression report that adult-born cells can remain comparatively excitable for several months, which challenges the view that their contribution is confined to their first few weeks (Fölsz et al., 2023).

2.4 Functional roles of adult-born granule cells

The two views of what adult-born cells do, set out in Chapter 1, differ in whether the cells are treated as part of the code or as something acting on it. The encoding view is supported by the observation that hyperexcitable DG cells are preferentially recruited into engrams, so that cells in the critical window are more readily incorporated into any memory trace than their mature neighbours, and by reports that they are activated during both encoding and retrieval (Fölsz et al., 2023; Piatti et al., 2013). The modulatory view rests instead on what these cells do to the rest of the circuit, and it is the one for which circuit-level evidence has accumulated most recently (Aimone, 2016; Fölsz et al., 2023; McHugh et al., 2022).

Three mechanisms have been proposed for that modulatory action, and they are not exclusive. The first is competition for synaptic partners. Electron microscopy and live imaging of synaptogenesis on adult-born cells showed that dendritic filopodia are most abundant early in maturation and that the tip of every filopodium examined lay within 200 nm of a pre-existing bouton already synapsing on another neuron, with new spines contacting multiple-synapse boutons preferentially; the connectivity of these cells continued to change for at least two months (Toni et al., 2007). New cells therefore insert themselves into established connections rather than building fresh ones, and later displace mGC axons entirely (Fölsz et al., 2023; Toni et al., 2007).

The second is inhibition of mGCs, both directly and through interneurons. Here the sign of the interaction depends on which input stream drives the adult-born cell: those receiving lateral entorhinal input inhibit mGCs, whereas those receiving medial entorhinal input excite them (Luna et al., 2019). Evidence that the net effect on the surrounding network is inhibitory comes from removing the cells: after focal irradiation or pharmacogenetic ablation of the young adult-born population, perforant-path evoked responses in the DG were reduced, but spontaneous gamma-frequency bursts in the DG and hilus grew markedly in amplitude and dentate firing became more tightly synchronized to them, which led the authors to conclude that young neurons normally inhibit or destabilize recurrent activity in the DG and hilus (Lacefield et al., 2012).

The third is inhibition reaching beyond the DG. Adult-born cells engage hippocampal

interneurons that inhibit neighbouring mGCs and also exert inhibition on CA3 and CA1, with the reported consequence being an overall sparsification of population firing across the hippocampus (Fölsz et al., 2023; McHugh et al., 2022). The clearest evidence that the effect crosses regions is that suppressing adult neurogenesis, either chemically or genetically, impairs the population-based coding of similar contexts in *CA3* while leaving the coding of dissimilar contexts intact (Niibori et al., 2012). That the deficit appears downstream of the DG, and only for similar contexts, is the observation this work sets out to explain mechanistically.

The behavioural evidence is broadly consistent with a role in discrimination but is not uniform. Increasing neurogenesis is sufficient to improve behavioural pattern separation (Sahay et al., 2011), and cells in the critical window are required for tasks that demand the separation of highly similar contexts while appearing dispensable for learning a single location in the water maze (Fölsz et al., 2023).

A separate proposal is that adult-born cells provide a time stamp, linking memories encoded close together in time while separating remote ones, on the grounds that the population available for recruitment changes as cells are added and mature (Aimone et al., 2009). This has some *in vivo* support: recordings from the DG identified cells whose activity was selective for a single context when the exposures to the two contexts were widely separated in time, with that selectivity attenuating as the exposures were brought closer together and attenuating further when neurogenesis was reduced, which the authors interpret as a temporally orthogonalizing code in the DG (Rangel et al., 2014). Support for the idea is nonetheless mixed, since little overlap has been found between the adult-born cells active during encoding and during retrieval of contextual fear memories, which has been taken to suggest that these cells respond to behavioural state rather than to particular events (Fölsz et al., 2023). Beyond encoding, adult-born cells have been implicated in later stages of memory processing, including consolidation during sleep, and in the gradual degradation of old connections that promotes generalization and eventually forgetting (Chavan et al., 2025; Fölsz et al., 2023).

Consistent with the latter, McHugh et al. (2022) showed *in vivo* that immature adult-born GCs carry little spatial information, especially when compared to mGCs; this study also showed that briefly activating those iGCs recruits inhibitory interneurons throughout the hippocampus

and increases population sparsity in both the DG and CA3, pointing to a circuit-level mechanism by which a small population of new cells could regulate activity well beyond the DG. Similarly, Ko et al. (2025) showed that adult neurogenesis drives the reorganization of hippocampal engram circuitry as memories are consolidated, remodeling the DG-CA3-CA1 network so that its representations generalize over time. These results show that iGCs have important effects across the hippocampus network, not being restricted to a local function in the DG only.

2.5 Computational models of the hippocampus and of adult neurogenesis

2.5.1 Neuron and synapse models

Any network model has to choose how much of single-cell biophysics to keep. The integrate-and-fire abstraction, which reduces a neuron to a leaky capacitor with a threshold, dates to the beginning of the last century (L. Abbott, 1999; Lapicque, 1907) and remains cheap enough for large networks, but it discards the mechanisms that generate the action potential. At the other extreme, the conductance-based formulation derived from the squid axon (Hodgkin & Huxley, 1952a) reproduces those mechanisms explicitly at a substantially higher computational cost. Between the two sit reduced models with a small number of parameters that nonetheless recover a wide range of firing patterns, of which the Izhikevich model is the one adopted here (E. Izhikevich, 2003; E. M. Izhikevich, 2006). How faithfully such reduced models reproduce the spiking of real neurons has been assessed systematically, and generalized leaky integrate-and-fire variants can classify multiple neuron types from their responses (Gerstner & Naud, 2009; Teeter et al., 2018). The relevant synaptic phenomena for this work are the short-term ones, in which transmission is depressed by the depletion of releasable resources and facilitated by residual calcium, and so the Tsodyks–Markram model was used (Tsodyks et al., 1998; Zucker & Regehr, 2002). The particular equations used are stated in Chapter 3.

2.5.2 Network models of the dentate gyrus and CA3

Models of the DG span the same range of detail, from abstract treatments of its coding properties through phenomenological circuit models to fully conductance-based

reconstructions (Aimone, 2016). Which features prove necessary depends on the question: the dendritic morphology of the GCs, for instance, turns out to control DG sparsity and separation directly and cannot simply be abstracted away (Chavlis et al., 2017). Models of CA3 range from detailed and simplified pyramidal cell reconstructions (Migliore et al., 2019) to full-scale data-driven networks in which cell assemblies form and are retrieved (Kopsick et al., 2024); broader surveys of hippocampal spiking models and of the microcircuit data that constrain them are available (Cutsuridis, 2019; Sutton & Ascoli, 2021), as are neuromorphic implementations of the circuit (Casanueva-Morato et al., 2023).

Two practical constraints shape this literature. The first is that assembling a biologically constrained network requires a consolidated source of neuron and synapse parameters, which is the role played by knowledge bases such as Hippocampome.org (Wheeler et al., 2023). The second is computational: simulating large spiking networks on general-purpose hardware has known limits (Zenke & Gerstner, 2014), which is one reason models are built at reduced scale and with efficient simulators (Stimberg et al., 2019). It is also worth noting an anatomical fact that constrains any DG-CA3 model: the mossy fibre forms giant multisynapse terminals on its CA3 targets, yet each fibre contacts only on the order of a dozen pyramidal cells, making it one of the sparsest connections in the brain, and its strength is offset by the heavy tonic inhibition the GCs receive (Aimone, 2016).

2.5.3 Models of adult neurogenesis

Computational work on DG neurogenesis has been organized at three scales, namely the individual new neuron, the DG as a circuit, and the hippocampus as a loop, and the questions asked differ at each (Aimone, 2016; Aimone & Gage, 2011). At the level of the single cell, the excitability of the immature neuron has been attributed to a weaker and even depolarizing response to GABA in the first weeks, to a distinct complement of ion channels giving a higher input resistance, and to the reduced synaptic input the cell receives; modelling indicates that the reduced input alone is sufficient to bias an immature GC towards firing more often than a mature one (Aimone, 2016). That last point matters for the present work, because reduced afferent input is the mechanism by which immaturity is represented in the model described in Chapter 3.

At the level of the whole hippocampal loop, the earliest models incorporated new neurons into a four-layer circuit and found that turnover was necessary for effective coding of representationally similar inputs, while later work treating neurogenesis as an additive rather than a replacement process found that the new cells came to represent much of the response to novel inputs and, in that configuration, showed signs of impaired pattern separation (Aimone, 2016). Incorporating neurogenesis into the most detailed models is difficult for a simple reason: neurogenesis takes place over weeks, whereas conductance-based models are integrated at sub-millisecond steps, so the two require timescales that are difficult to combine in one simulation (Aimone, 2016).

A more recent theoretical treatment attributes the functional switch to the GABAergic input onto the maturing cell. While that input is depolarizing, cooperativity drives synaptic growth so that the newborn cell becomes responsive to stimuli resembling those that already activate mature cells; once the input becomes inhibitory, competition pushes its synaptic configuration towards stimuli unlike those already stored. The model thereby accounts both for how adult-born cells integrate into an existing network and for why they promote the separation of similar but not of distinct patterns (Gozel & Gerstner, 2021), which is the same asymmetry found experimentally in CA3 (Niibori et al., 2012).

Reviews of this literature have noted that models agree that neurogenesis and pattern separation are related but disagree about what is being separated, and that several distinct circuit mechanisms could produce the same behavioural outcome (Aimone, 2016; Wang et al., 2024). Three candidates have been set out explicitly. The first is a population effect, in which the new cells themselves represent new information and their distinct connections to CA3 reduce interference between memories. The second is that new cells restrain activity in the surrounding DG, making the activity of the mature population sparser and thereby more orthogonal; the loss-of-function evidence for this is the increase in dentate and hilar gamma bursting that follows ablation of the young population (Lacefield et al., 2012). The third is that they differentially engage CA3 interneurons, recruiting feedforward inhibition in CA3 and priming the recurrent network (Aimone, 2016). The first is an encoding mechanism and the other two are circuit mechanisms that act on the mature population; distinguishing between them requires a model

that includes CA3, which the studies reviewed below do not.

In a conductance-based DG model (Kim & Lim, 2024b), the large population of mGCs performed excellent pattern separation, while the minority of iGCs served as pattern integrators. The presence of iGCs deteriorated the DG's pattern separation capacity compared to a DG composed solely of mGCs; however, the authors postulate that this mixed coding of pattern integration and separation could increase the capacity for memory storage and retrieval, a hypothesis they were unable to test since only the DG was modeled, without CA1 or CA3. Interestingly, these results contradict experimental studies demonstrating that an increase in neurogenesis enhances pattern separation (Sahay et al., 2011), which may indicate that computational models of the hippocampus have not yet fully captured the computations it performs.

In contrast, the work of Yang et al. (2025) employed a DG model to investigate the dynamic impact of neurogenesis on pattern separation. Their results indicate that the general effect of iGCs on GCs is inhibitory, which increases network sparsity and consequently improves pattern separation, providing computational evidence for the "indirect coding" hypothesis. The authors also investigated how iGC characteristics during maturation, such as decreasing activity and increasing synaptic plasticity, dynamically affect pattern separation. The study further suggests that the presence of iGCs increases the adaptability of the DG network to different input frequencies, improving the robustness of pattern separation.

The model of Aimone et al. (2009) was implemented at an intermediate level of biological plausibility[1], which allowed for the investigation of iGCs throughout their entire maturation over time. By modeling the maturation timeline of this iGC population, the model showed that iGCs preferentially encoded information received when they were young, thus exhibiting the role of temporal integrators. This model was corroborated by experimental studies (Berdugo-Vega et al., 2023), but did not analyze the implications for other areas such as CA1 and CA3, modeling only the DG.

[1] This model is considered to be at an intermediate level of biological plausibility because it does not model the electrical dynamics of individual neurons through conductance-based models, as in this project, but instead uses a digitized spike-rate model with a time step of 25 ms.

3 Materials and Methods

3.1 DG-CA3 neural network model

Based primarily on the models of Chavlis et al. (2017), Kim and Lim (2024a), Kopsick et al. (2024), and Yang et al. (2025), the DG-CA3 network architecture was modeled as shown in Figures 1 and 2 at a scale of $\frac{1}{500}$ of the rat hippocampus, with the number of neurons per population described in Table 2. Neurons were modeled simply, with a single compartment, but with high biological plausibility, using neuron and synapse models and cell count and connectivity data adapted from Hippocampome.org, an open-access knowledge base that compiles multiple data sources on the hippocampal formation (Wheeler et al., 2023).

The model included the core excitatory and inhibitory cell types of the DG-CA3 network underlying pattern separation and completion: in the DG, GCs, MCs, BCs and HIPPs; in CA3, pCA3 and iCA3 cells. Although both regions contain many additional interneuron subtypes, several of which are also characterized in Hippocampome.org (Wheeler et al., 2023), this reduced set has been shown in previous computational models to be sufficient to support pattern separation (Chavlis et al., 2017; Kim & Lim, 2024b; Yang et al., 2025). Synaptic connectivity between cell types and their connection probabilities were likewise derived from multiple sources and constrained to biologically plausible ranges. After the parameters had been gathered from Hippocampome.org, and the network had been assembled, a subset of neuron and synapse parameters, together with the selected connection probabilities, was adjusted within biologically reasonable bounds to reproduce the *in vivo* firing rate and population sparsity associated with each cell type (Table 1).

The network input is composed of cells from the entorhinal cortex (EC), with a total of $N_{EC} = 400$ neurons (Amaral et al., 1990; Kim & Lim, 2024a). In each simulation, the EC as a whole presents a specific pattern, where each pattern is represented by an active subpopulation of 10% of EC neurons (McNaughton et al., 1991). Inactive neurons not belonging to the pattern do not spike during the simulation, whereas active neurons spike according to a Poisson distribution with a spike rate of 40 Hz. EC neurons project their axons via the perforant path to neurons with

Figure 1 – DG-CA3 network architecture. Inhibitory synapses are represented by circles and excitatory ones by arrows. The network input is composed of cells from the entorhinal cortex (EC). The dentate gyrus (DG) has the perforant path and the dendrites of its cells in the molecular layer; the granular layer is composed of immature and mature Granule Cells (iGCs and mGCs, respectively) and Basket Cells (BCs); the hilus is composed of Mossy Cells (MCs) and Hilar Perforant Path-associated cells (HIPP). The CA3 is composed of the pyramidal cells, labelled **P**, and the inhibitory interneurons, labelled **I**, referred to in the text as pCA3 and iCA3, respectively. The CA1 box marks the efferent target of the pCA3 cells and is not part of the model.

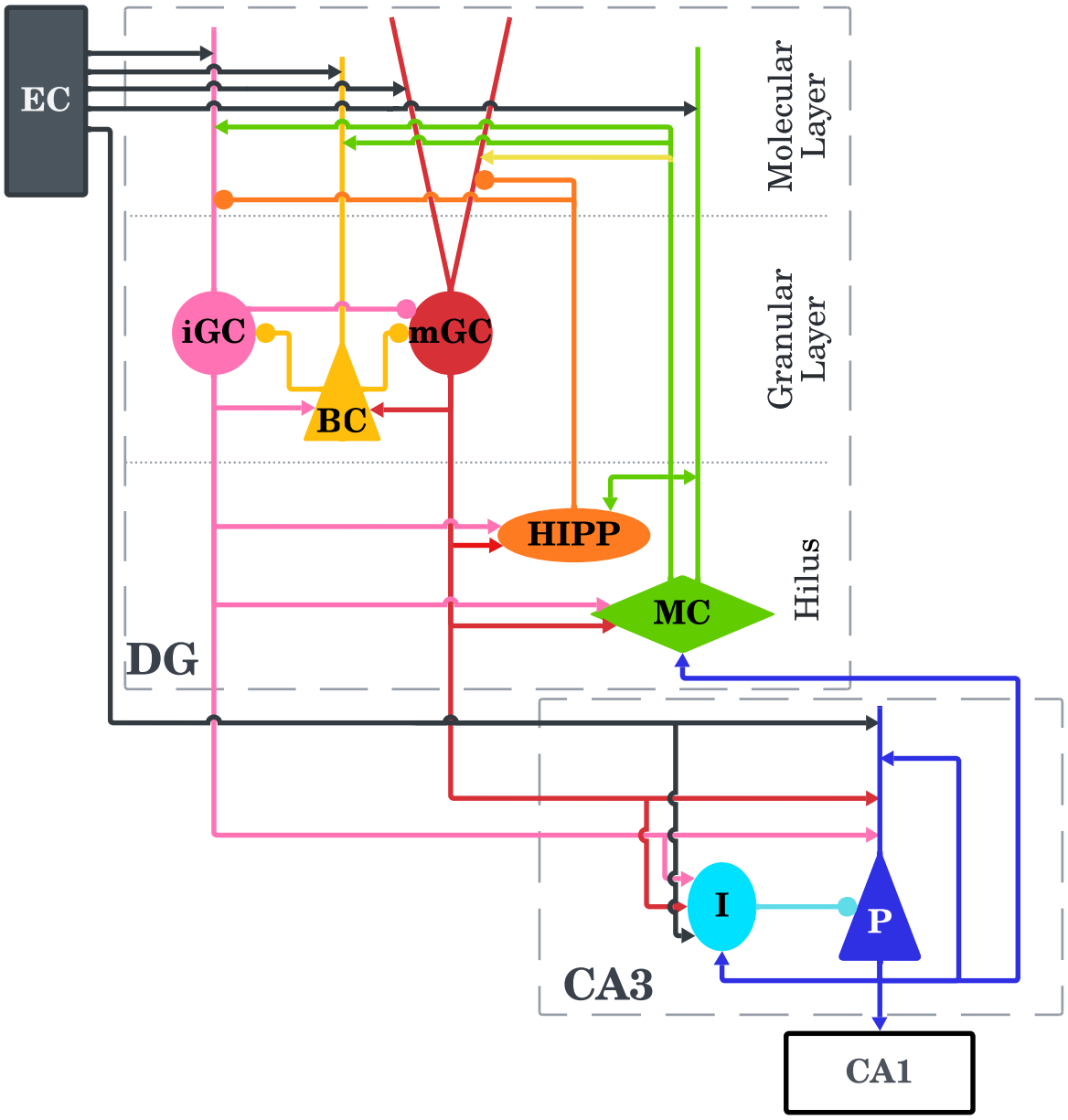
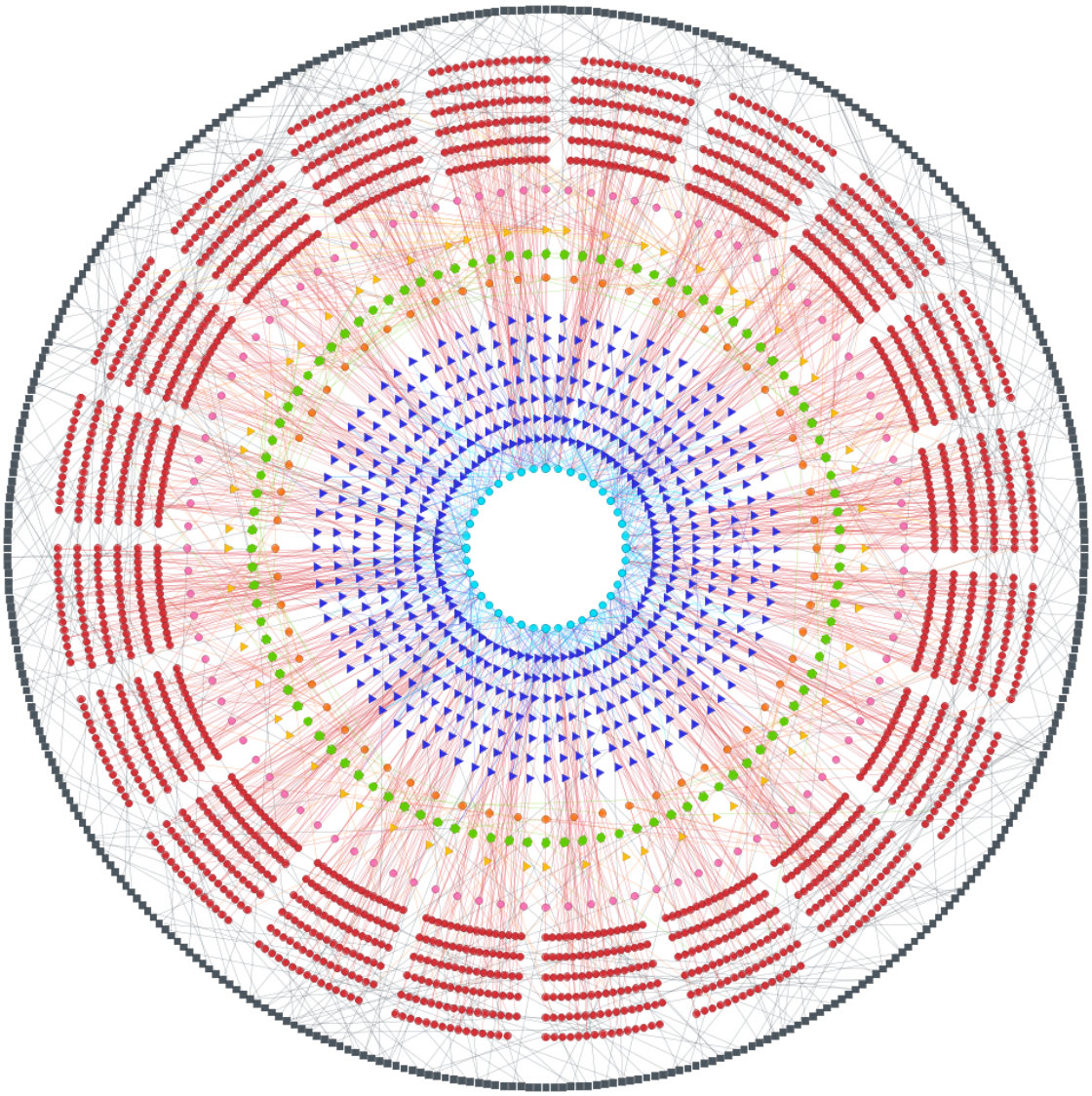


Figure 2 – Ring graph of the DG-CA3 network. Each node represents a neuron in the model, with the same color scheme as Figure 1 for each neuron. Each edge represents a synapse, colored according to the presynaptic neuron, although here only a subset of 2% of the synapses are represented. The EC is represented by the outermost black circle; the lamellar organization of the DG can be seen with mGCs and iGCs (red and pink, respectively) and pCA3 cells at the center in teal. In total, the graph contains 3,280 nodes and 266,596 edges.



Cell	*In Vivo* Firing Rate (Hz)	Reference	Model Firing Rate (Hz)	*In Vivo* Activation	Reference	Model Activation
Mature granule	1.2 (0.56–2.38)	McHugh et al. (2022)	1.08	2–5%	Galloni et al. (2022) and Hainmueller & Bartos (2018)	8.0%
	0.49	Diamantaki et al. (2016)		9%	GoodSmith et al. (2017)	
	0.031 ± 0.096	Zhang et al. (2020)				
	0.4	Wheeler et al. (2023)				
	0.96 ± 0.19	GoodSmith et al. (2019)				
	≈ 0.2	Senzai & Gyorgy Buzsaki (2016)				
	72 ± 8^{max}	Lübke et al. (1998)				
Immature granule	2.3 (1.6–6.7)	McHugh et al. (2022)	2.96			63.8%
Mossy	0.84	Wheeler et al. (2023)	3.45	≈ 100%	Danielson et al. (2017)	96.7%
	1.05 ± 0.07	GoodSmith et al. (2019)		88%	GoodSmith et al. (2017)	
	≈ 0.8	Senzai & Gyorgy Buzsaki (2016)				
	50 ± 6^{max}	Lübke et al. (1998)				
HIPP	111.4 ± 10.2^{max}	Yuan et al. (2017)	6.38			100%
Basket	13.6 ± 0.8	Li et al. (2022)	17.35			100%
	230 ± 15^{max}	Lübke et al. (1998)				
CA3 pyramidal	0.2	Mizuseki & Buzsáki (2013)	1.17	29%	GoodSmith et al. (2017)	30.0%
	0.5	Kay et al. (2016)		20–30%	Galloni et al. (2022)	
	0.72 ± 0.51	Lasztóczi et al. (2011)				
	CA3a: 0.4; CA3b: 0.3	Oliva et al. (2016)				
	1.2	McHugh et al. (2022)				
	0.92 ± 0.06	GoodSmith et al. (2019)				
	≈ 0.6	Senzai & Gyorgy Buzsaki (2016)				
CA3 inhibitory	20 ± 7	Tukker et al. (2013)	6.51			65.4%
	17 ± 7	Lapray et al. (2012)				
	8.2 ± 5.6	Leão et al. (2012)				

Table 1 – *In vivo* firing rates and activation (the fraction of active cells) for each cell type, alongside model values. *In vivo* firing rates are mean ± SD. ^{max} maximum firing rate evoked by depolarizing current injection *in vitro* (mean ± SEM for the HIPP cells). ≈ approximate value read from figure. Model firing rates are the mean over active cells and model activation the fraction of active cells; both are from the control network, except the immature granule cells (absent in the control), which are from the neurogenesis model at 50% EC→iGC excitability.

dendrites in the molecular layer of the DG: Granule Cells (GCs), Mossy Cells (MCs) (Scharfman & Myers, 2013), and Basket Cells (BCs), as well as to CA3 neurons.

GCs are subdivided into two subpopulations: mature Granule Cells (mGCs) and immature Granule Cells (iGCs), representing GCs generated by adult neurogenesis at their characteristic electrophysiological phase of 4–7 weeks of age (Aimone et al., 2014). In total, the network is composed of $N_{GC} = 2000$ GCs ($\frac{1}{500}$ of the 10^6 rat granule cells) (West et al., 1991), with 5% of them being iGCs (Cameron & Mckay, 2001), giving $N_{mGC} = 1900$ and $N_{iGC} = 100$. Following the lamellar organization of the DG (Sloviter & Lømo, 2012), GCs are distributed across 20 lamellae, with 100 cells per lamella. Each GC connects with the BCs and CA3 neurons of the same lamella, and also forms random, non-lamellar excitatory connections with HIPPs and MCs.

iGCs possess distinct electrophysiological characteristics from the mGCs that render them more excitable than mGCs. Furthermore, to simulate their not-yet-complete circuit integration, iGCs receive EC afferent inputs at only a fraction $x_{EC\rightarrow iGC}$ of the connection probability between

the EC and mGCs, and also receive reduced inhibitory inputs from BCs and HIPPs (Table 4). Additionally, some iGCs are known to also directly inhibit mGCs (Luna et al., 2019), so the model includes an inhibitory synapse from iGCs onto mGCs. The MC→iGC projection is left unscaled, since MC axons terminate on the proximal dendrites in the inner molecular layer while the EC afferents terminate distally (Amaral et al., 2007), and the dendrites of an adult-born GC extend outwards as it matures (Zhao et al., 2006), reaching the MC input before the EC one. Because Hippocampome.org (Wheeler et al., 2023) does not characterize iGCs, six of the nine iGC parameters were fitted to electrophysiological data reported by Espósito et al. (2005). Parameter fitting was performed using the Nelder and Mead (1965) simplex method by minimizing the squared error between the model output and the number and timing of the spikes evoked by a range of depolarizing current steps. The membrane capacitance, resting potential and spike peak voltage were held fixed (Table 3).

BCs ($N_{BC} = 60$) are inhibitory cells that ensure sparse GC activation through winner-take-all competition among them (Chavlis et al., 2017; Coultrip et al., 1992; Kim & Lim, 2024a), where only the most active GCs in a lamella remain active, inhibiting the rest. HIPPs ($N_{HIPP} = 60$) also contribute to the sparsity of GC activation, with a more global inhibition acting on all GCs in a lamella. MCs ($N_{MC} = 100$) are excitatory hilar cells that receive excitation from GCs and project back to GCs, BCs, HIPPs and other MCs via cross-lamellar connections. Although these excitatory MC-to-GC projections exist, their net effect is generally inhibitory through the control of BCs and HIPPs (Myers & Scharfman, 2009; Scharfman & Myers, 2013).

The HIPP parameters in Hippocampome.org (Wheeler et al., 2023) model the "rebound burst spiking" characteristic, in which the neuron produces a burst of spikes following the end of a hyperpolarization period. This model is incompatible with the network, as large hyperpolarizations lead to infinite spike bursts. Therefore, the parameters used for HIPPs were taken from Modak and Chakravarthy (2018).

CA3, unlike the DG, does not follow a lamellar structure in the model, as it forms a far more integrative neural network (Pak et al., 2022; Watson et al., 2025): its internal connections, both among pCA3 cells and between the pCA3 and iCA3 populations, are formed at random rather than within lamellae. The CA3 model comprises 600 pyramidal cells ($N_{pCA3} = 600$)

and 60 inhibitory cells (N_{iCA3} = 60), modeled according to the physiological data for CA3 basket cells (Wheeler et al., 2023) so as to simplify the variety of inhibitory neurons present in CA3 (Kopsick et al., 2024). Both neuronal populations receive afferent inputs from the EC and from GCs, with iCA3 cells inhibiting pCA3 cells. pCA3 cells excite iCA3 cells and, importantly, form a recurrent network by connecting with other pCA3 cells, a fundamental characteristic for the autoassociation and pattern completion functions of CA3 (Kopsick et al., 2024; Rolls, 2013). pCA3 cells also send feedback projections to the DG, connecting with MCs in the model, a process that contributes to pattern separation (Myers & Scharfman, 2011). Although CA3 has no lamellar organization of its own, this backprojection does follow the lamellae of the DG: each pCA3 cell contacts only MCs of its own lamella. Together with the lamellar GC→CA3 projections, this closes the DG-CA3 loop within a lamella.

All simulations were performed with Brian2 (Stimberg et al., 2019), using the 4th-order Runge-Kutta method with a fixed time step of 0.1 ms (Butcher, 1996). Neuron and synapse parameters can be found in Tables 3 and 4, respectively. The duration of a single simulation for one input pattern is 300 ms for the network to reach a steady state, with the pattern recording phase occurring subsequently for 1000 ms. Synaptic connections are kept fixed across all simulations.

Cell	EC	mGC	iGC	MC	HIPP	BC	pCA3	iCA3
N	400	1900	100	100	60	60	600	60

Table 2 – Neuronal population size (N).

3.2 Neuron model

Neurons were modeled according to the 9-parameter Izhikevich neuron model (E. M. Izhikevich, 2006, chapter 8) with a single compartment, without considering dendrites or axons. This model was chosen in this work and by Wheeler et al. (2023) for being capable of capturing the dynamic behavior of neurons across a wide variety of conditions with biological plausibility, comparable to the Hodgkin-Huxley model (Hodgkin & Huxley, 1952b), while presenting a simpler and computationally more efficient mathematical formulation. The spike traces of

neurons can be seen in Figure 3. The Izhikevich neuron model is described by the following equations:

$$C_m \frac{dV_m}{dt} = k(V_m - V_r)(V_m - V_t) - u + I \tag{3.1}$$

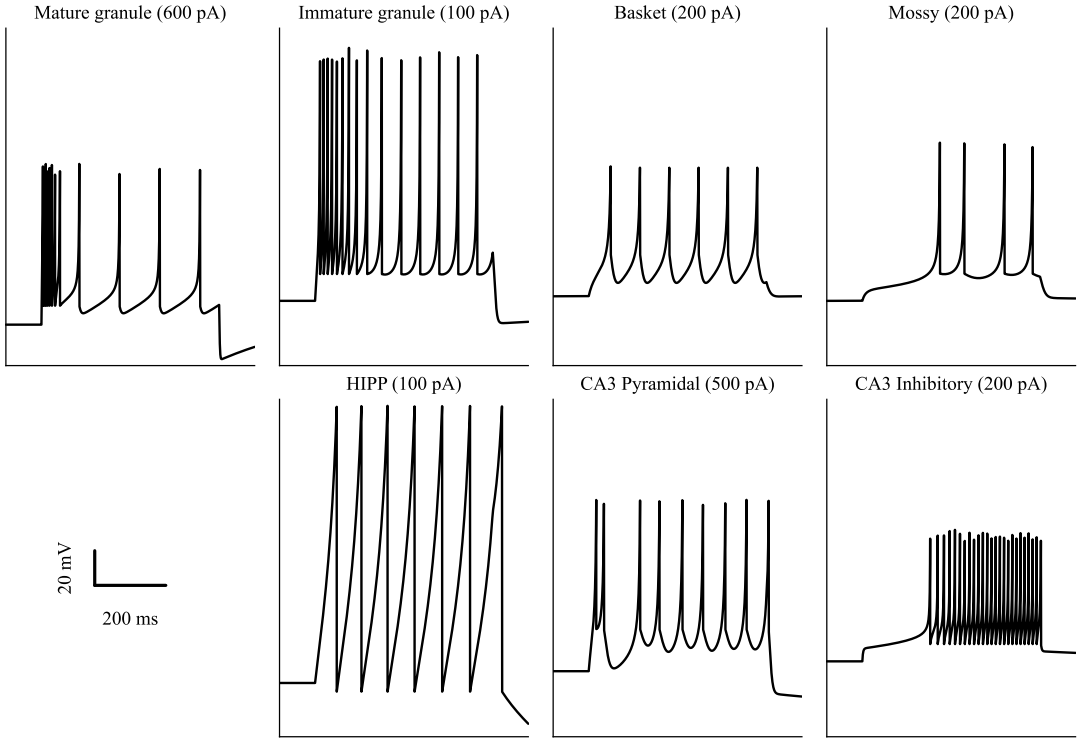
$$\frac{du}{dt} = a[b(V_m - V_r) - u] \tag{3.2}$$

Where V_m is the membrane potential, u is the recovery variable, C_m is the membrane capacitance, V_r is the resting potential, V_t is the threshold potential, I is the total current flowing into the neuron, and k, a, and b are constants that define the dynamic characteristics of the neuron. In addition to the differential equations above, which define the temporal evolution of the membrane potential and recovery variable, the Izhikevich neuron model also includes a rule for action potential generation, defined by Equation 3.3.

$$\text{if } V_m \geq V_{\text{peak}}, \quad \begin{cases} V_m \leftarrow V_{\text{min}} \\ u \leftarrow u + d \end{cases} \tag{3.3}$$

When the membrane potential reaches the peak value V_{peak}, an action potential is generated, the membrane potential is reset to the after-spike potential V_{min}, and the recovery variable u is incremented by d, making subsequent spike generation more difficult.

Figure 3 – Spike traces of neurons under constant current as indicated by each subfigure title.



Cell	k (nS/mV)	a (ms^{-1})	b (nS)	d (pA)	C_m (pF)	V_r (mV)	V_t (mV)	V_{min} (mV)	V_{peak} (mV)
mGC	0.45	0.003	24.48	50	38	-77.4	-44.9	-66.47	15.49
iGC	0.139	0.002	-1.877	12.149	24.6	-63.66	-38.41	-48.2	83.5
MC	1.5	0.004	-20.84	117	258	-63.67	-37.11	-47.98	28.29
HIPP	0.01	0.004	-2	40.52	58.7	-70	-50	-75	90
BC	0.81	0.097	1.89	553	208	-61.02	-37.84	-36.23	14.08
pCA3	0.79	0.008	-10	588	186	-63.2	-33.6	-38.87	35.86
iCA3	1	0.004	-18.26	12	45	-57.51	-23.38	-47.56	18.45

Table 3 – Izhikevich model parameters per cell type.

3.3 Synapse model

The synapse model, like the neuron model, was defined based on Hippocampome.org (Wheeler et al., 2023), following the formulation of Senn et al. (2001) and Mongillo et al. (2008). This model captures short-term plasticity, whether short-term

depression, caused by neurotransmitter depletion, or short-term potentiation, caused by calcium accumulation, both occurring on the scale of tenths of seconds. Each synapse has 5 parameters (described in Table 4): the maximum synaptic conductance in the absence of any synaptic resource depletion g, the proportion of resources utilized per spike U_{se}, the synaptic current decay time constant τ_d, the facilitation time constant τ_f, and the resource recovery time constant τ_r (Moradi et al., 2022). Another parameter shown in Table 4 is the connection probability P. Although Hippocampome.org provides data on connection probabilities between neuronal populations *in vivo*, the reduced scale of the network necessitated increasing these probabilities so that network activity could be maintained while sparse activity was preserved.

The model is described by three state variables: synaptic resource utilization (U), resource recovery (R), and the percentage of resources in an active state (A). Initially, $U_{t_0} = 0$, $R_{t_0} = 1$, and $A_{t_0} = 0$, since all resources are available for use. The temporal evolution of these variables is governed by the following system of differential equations:

$$\frac{dU}{dt} = \frac{-U}{\tau_f} + U_{se}(1 - U_{-})\delta(\Delta t_i) \tag{3.4}$$

$$\frac{dR}{dt} = \frac{1 - R - A}{\tau_r} - U_{+}R_{-}\delta(\Delta t_i) \tag{3.5}$$

$$\frac{dA}{dt} = \frac{-A}{\tau_d} + U_{+}R_{-}\delta(\Delta t_i) \tag{3.6}$$

where δ is the Dirac delta function, yielding 1 only when $\Delta t_i = t - t_i = 0$, i.e., only at time t corresponding to the synaptic event time t_i. U_{+} corresponds to the value of U immediately after the synaptic event, while R_{-} corresponds to the value of R immediately before the same event.

From these equations, the synaptic current is given by:

$$I_{syn} = s \cdot A \cdot g \cdot (V_m - E) \tag{3.7}$$

where V_m is the postsynaptic membrane potential, E is the synaptic reversal potential, for inhibitory and excitatory synapses, respectively, $E_{inh} = -86\,\text{mV}$ and $E_{exc} = 0\,\text{mV}$, and s is

a scaling constant defined as $s = 10$ for all synapses. This scaling constant is necessary due to the reduced scale of the network, since, given the low number of synapses in the model compared to the rat hippocampus, the entire network would fall silent without this scaling. All synaptic currents I_{syn} of a neuron are summed and affect the membrane potential V_m according to Equation 3.1.

3.4 Pattern separation

The methodology for quantifying pattern separation was based on that used by Kim and Lim (2024b). To analyze pattern separation under different levels of input pattern similarity and the impact of adult neurogenesis, 11 different model variants were tested: one control without adult neurogenesis ($N_{mGC} = 2000$) and 10 others with neurogenesis at different EC→iGC excitability levels ($N_{mGC} = 1900$ and $N_{iGC} = 100$), as illustrated in Figure 4. The excitability level is expressed as a percentage, where 100% corresponds to iGCs having the same EC-GC connection probability as mGCs; the intent is to simulate the incomplete circuit integration of iGCs, whose dendrites are not yet fully developed. Rather than committing to a single value, the whole range from 10% to 100% in increments of 10% was swept, since adult-born GCs acquire their entorhinal input gradually as they mature (Aimone et al., 2014; Espósito et al., 2005), so no single value represents the immature population. For each model, 30 sets of 10 input patterns are generated. Each set consists of one randomly generated original pattern and 9 patterns similar to it, ranging from 90% to 10% similarity. An input pattern consists of a binary vector of size $N_{EC} = 400$, representing the activation state of each EC neuron, in which active neurons spike according to a Poisson distribution at a frequency of 40 Hz. Every input pattern has exactly 10% active neurons. To generate patterns with different levels of similarity, a random fraction of X% of the active neurons is kept active, while the remaining $(100 - X)$% are deactivated and an equal number of initially inactive neurons is activated, to maintain the 10% activity level (40 active neurons).

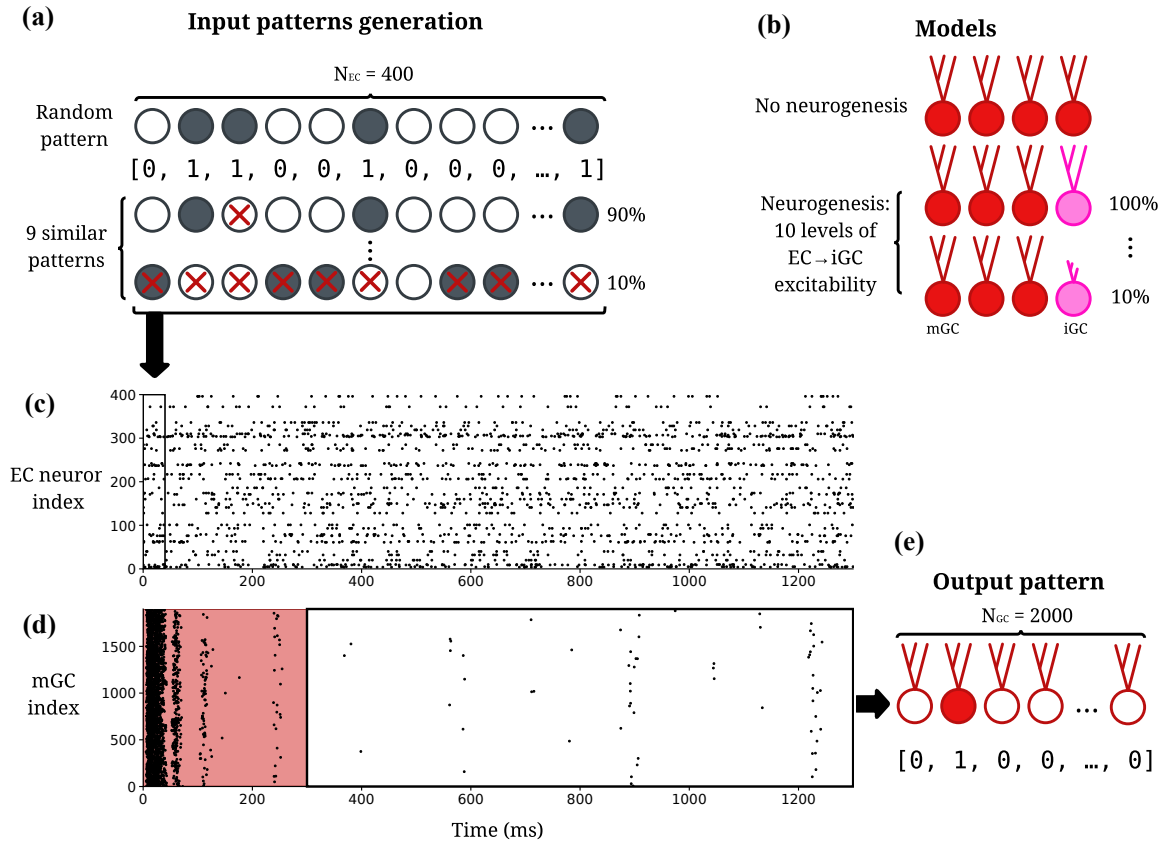
Each pattern is simulated for 1300 ms of simulation time, with the first 300 ms discarded while the network stabilizes. The output pattern of a neuronal population, whether GCs or pCA3 cells, consists of a binary vector with value 1 for the index of each neuron that spiked at least

Presynaptic	Postsynaptic	Connection	P (%)	g (nS)	τ_d (ms)	τ_r (ms)	τ_f (ms)	U
EC	mGC	Random	8	1.655	5.333	266.239	18.714	0.27
EC	iGC	Random	0.8–8*	1.655	5.333	266.239	18.714	0.27
EC	MC	Random	20	1.522	4.671	319.835	57.766	0.204
EC	BC	Random	25	1.406	3.849	144.415	48.2	0.214
EC	pCA3	Random	12	1.5	6.55	258.318	53.478	0.184
EC	iCA3	Random	12	2	3.602	257.468	35.904	0.21
mGC	MC	Random	40	2.713	5.347	98.583	27.479	0.151
mGC	HIPP	Random	10	1.805	5.181	362.814	48.986	0.15
mGC	BC	Lamellar	100	4.258	5.566	31.265	4.278	0.497
mGC	pCA3	Lamellar	80	1.5	6.657	278.286	78.584	0.155
mGC	iCA3	Lamellar	50	3	3.915	218.934	43.274	0.176
iGC	mGC	Random	10	0.524	3.915	218.934	43.274	0.176
iGC	MC	Random	40	2.713	5.347	98.583	27.479	0.151
iGC	HIPP	Random	10	1.805	5.181	362.814	48.986	0.15
iGC	BC	Lamellar	100	4.258	5.566	31.265	4.278	0.497
iGC	pCA3	Lamellar	80	1.5	6.657	278.286	78.584	0.155
iGC	iCA3	Lamellar	50	3	3.915	218.934	43.274	0.176
MC	mGC	Cross-lamellar	0.2	1.894	5.357	166.162	20.224	0.304
MC	iGC	Cross-lamellar	0.2	1.894	5.357	166.162	20.224	0.304
MC	MC	Cross-lamellar	5	0.068	4.257	149.329	71.642	0.245
MC	HIPP	Cross-lamellar	50	2.376	4.824	358.431	54.872	0.181
MC	BC	Cross-lamellar	100	1.996	3.396	117.365	69.316	0.255
HIPP	mGC	Random	30	3.202	8.935	159.143	10.396	0.278
HIPP	iGC	Random	10	3.202	8.935	159.143	10.396	0.278
BC	mGC	Lamellar	100	4.351	3.543	103.876	12.347	0.332
BC	iGC	Lamellar	10	4.351	3.543	103.876	12.347	0.332
pCA3	pCA3	Random	2.5	0.9	9.516	278.258	27.513	0.172
pCA3	MC	Lamellar	10	2.035	4.297	359.116	40.457	0.236
pCA3	iCA3	Random	20	2	4.525	275.604	23.321	0.189
iCA3	pCA3	Random	30	2	7.793	416.282	20.63	0.203

Table 4 – Parameters of the synapses between neuronal populations. Random connections occur between all cells of both populations; lamellar connections occur between cells of the same lamella; cross-lamellar connections occur between cells of one lamella and all others. The connection probability P refers to the percentage of connections between neuronal populations according to the connection condition. * iGCs were simulated with multiple different EC→iGC excitability levels.

once during the 1000 ms interval[1].

Figure 4 – Diagram of the input and output pattern generation and recall protocol. **(a)** Input patterns are generated in sets of 10 patterns of size $N_{EC} = 400$; the first is generated randomly and the remaining ones are generated from it at different levels of similarity by randomly activating and deactivating neurons. **(b)** 11 model variants are simulated: one without neurogenesis and 10 with neurogenesis at different excitability levels, from 100% down to 10% excitability. **(c)** Spike trace of EC neurons during a simulation; each generated pattern corresponds to the activation of EC neurons, with active neurons spiking at a frequency of 40 Hz; each of the 11 models is exposed to 30 sets of 10 input patterns, totaling 300 simulations of 1300 ms each. **(d)** Spike trace of mGCs during a simulation; the first 300 ms are discarded while the network stabilizes; the remainder corresponds to the output pattern. **(e)** The output pattern, illustrated here for the mGC population, consists of a binary vector with value 1 for the index of each neuron that spiked at least once during the 1000 ms interval.



To characterize pattern separation, the overlap between the neural activity patterns at the input (entorhinal cortex cells) and output (DG Granule Cells or CA3 pyramidal cells) of the network is compared. A pattern is defined by a binary representation of size N, where N is the

[1] Computing pattern separation from graded spike counts instead of binary activation was also considered, but it yields nearly identical results (Pearson $r = 0.99$).

total number of neurons in a specific population, with neurons that spiked at least once during the simulation time interval (1000 ms) represented by 1 and those that did not by 0. For a pair of patterns $A^{(l)}$ and $B^{(l)}$ (where $l \in \{in, out\}$ for input and output, respectively), the distance between patterns $D_p^{(l)}$ is defined as:

$$D_p^{(l)} = \frac{O^{(l)}}{D_a^{(l)}} \tag{3.8}$$

In this equation, $O^{(l)}$ represents the degree of orthogonalization and $D_a^{(l)}$ the mean degree of activation of the two patterns. The mean degree of activation $D_a^{(l)}$ is the arithmetic mean of the degrees of activation of each pattern, $A^{(l)}$ and $B^{(l)}$:

$$D_a^{(l)} = \frac{D_a^{(A^{(l)})} + D_a^{(B^{(l)})}}{2} \tag{3.9}$$

The degree of activation of an individual pattern is the fraction of active neurons (represented by 1 in a binary encoding) in the pattern. The orthogonalization degree $O^{(l)}$, which measures the dissimilarity between patterns, is calculated from the Pearson correlation coefficient, $\rho^{(l)}$:

$$O^{(l)} = \frac{1 - \rho^{(l)}}{2} \tag{3.10}$$

Where $\rho^{(l)}$ is the Pearson correlation coefficient between patterns $A^{(l)}$ and $B^{(l)}$. Considering $\{a_i^{(l)}\}$ and $\{b_i^{(l)}\}$ $(i = 1, \ldots, N_l)$ as the binary representations of the state of the i-th cell in patterns $A^{(l)}$ and $B^{(l)}$ $(l \in \{in, out\})$, the Pearson correlation coefficient is given by:

$$\rho^{(l)} = \frac{\sum_{i=1}^{N_l} \Delta a_i^{(l)} \cdot \Delta b_i^{(l)}}{\sqrt{\sum_{i=1}^{N_l} \left(\Delta a_i^{(l)}\right)^2} \sqrt{\sum_{i=1}^{N_l} \left(\Delta b_i^{(l)}\right)^2}} \tag{3.11}$$

where $\Delta a_i^{(l)} = a_i^{(l)} - \langle a^{(l)} \rangle$ and $\Delta b_i^{(l)} = b_i^{(l)} - \langle b^{(l)} \rangle$. The notation $\langle \ldots \rangle$ indicates the population mean over all cells. The value of $\rho^{(l)}$ varies between -1 and 1, therefore $O^{(l)}$ varies between 0 and 1 (Equation 3.10).

From the input $(D_p^{(in)})$ and output $(D_p^{(out)})$ pattern distances, the degree of pattern separation, $\mathcal{S}_D$, is calculated as the ratio between them:

$$\mathcal{S}_D = \frac{D_p^{(out)}}{D_p^{(in)}} \tag{3.12}$$

A value of $\mathcal{S}_D > 1$ indicates that the output patterns are more distinct than the input patterns, characterizing pattern separation. Conversely, $\mathcal{S}_D < 1$ indicates pattern convergence, where the output patterns become more similar to each other.

3.5 Population sparsity

Population sparsity measures how unevenly spiking is distributed across a population, being high when activity is concentrated in a few cells and low when it is spread evenly. Following McHugh et al. (2022), it was quantified with two complementary indices computed on the per-cell spike counts accumulated over the recording window. For a population of N cells with spike count vector x sorted in ascending order ($x_1 \leq \cdots \leq x_N$), the Gini index (Gini, 1921) is:

$$S_{Gini} = \frac{\sum_{i=1}^{N} (2i - N - 1)\, x_i}{N \sum_{i=1}^{N} x_i} \tag{3.13}$$

where i is the rank of each cell in the sorted vector. As a cross-validation, the Hoyer measure (Hoyer, 2004) was also computed:

$$S_{Hoyer} = \frac{\sqrt{N} - \left(\sum_{i=1}^{N} |x_i|\right) / \sqrt{\sum_{i=1}^{N} x_i^2}}{\sqrt{N} - 1} \tag{3.14}$$

Both indices range over $[0, 1]$, with larger values denoting sparser activity. They were computed for the mGC, iGC and pCA3 populations and for the combined GC population (mGCs and iGCs together), and averaged over all patterns and trials, with the standard error of the mean (SEM) reported as the error.

3.6 iGC activation and downstream circuit effects

McHugh et al. (2022) showed *in vivo* that briefly activating immature (4–7 weeks) adult-born GCs recruits hippocampal interneurons and increases population sparsity in the DG and CA3. To test whether the model reproduces these circuit effects, iGCs were activated

directly through current injection and the response of every population was measured. The same EC→iGC excitability variants as in the pattern-separation experiment were used, each driven by an identical EC input.

In each run, a random half of the iGCs (50 of the 100) received a depolarizing current of 0.5 nA for 5 ms, delivered 400 ms into the recording window, once network activity had stabilized. This injected current I_{ext} enters the neuron as part of the total current I of the Izhikevich model (Equation 3.1):

$$I = I_{ext} + I_{syn} \tag{3.15}$$

where I_{syn} is the summed synaptic current (Equation 3.7) and $I_{ext} = 0.5$ nA during the pulse and zero otherwise. For each variant the protocol was repeated over 30 trials of 10 input patterns, and the resulting 300 runs were pooled for analysis.

The response of each population was quantified with a peri-stimulus time histogram (PSTH) of its firing rate, computed in 1.5 ms bins over the interval from 60 ms before to 60 ms after the pulse. In each bin the population firing rate is the pooled spike count divided by the number of cells, the number of runs and the bin width, and is expressed as a z-score relative to a pre-pulse baseline, taken as the binned population rate from the start of the recording window up to 60 ms before the pulse:

$$z(t) = \frac{r(t) - \mu_{base}}{\sigma_{base}} \tag{3.16}$$

where $r(t)$ is the population rate in the bin at time t relative to the pulse, and μ_{base} and σ_{base} are the mean and standard deviation of the baseline bins.

To relate the activation to sparsity, as McHugh et al. (2022) did *in vivo*, the Gini index (Equation 3.13) of each population was also compared, with a paired Wilcoxon signed-rank test over all runs, between the 50 ms windows immediately before and after the pulse.

3.7 Oscillation analysis

To characterize the rhythmicity evoked in BCs by the activation, the post-pulse population rate (5 ms to 60 ms after pulse onset, in 1 ms bins) was detrended by subtracting a Gaussian-smoothed envelope (σ = 8 ms), and the power spectrum of the residual was estimated by Welch's method (Welch, 1967). The peak frequency in the 20 Hz to 200 Hz band was taken as the dominant oscillation frequency (Figure 12).

4 Results

4.1 Baseline network activity

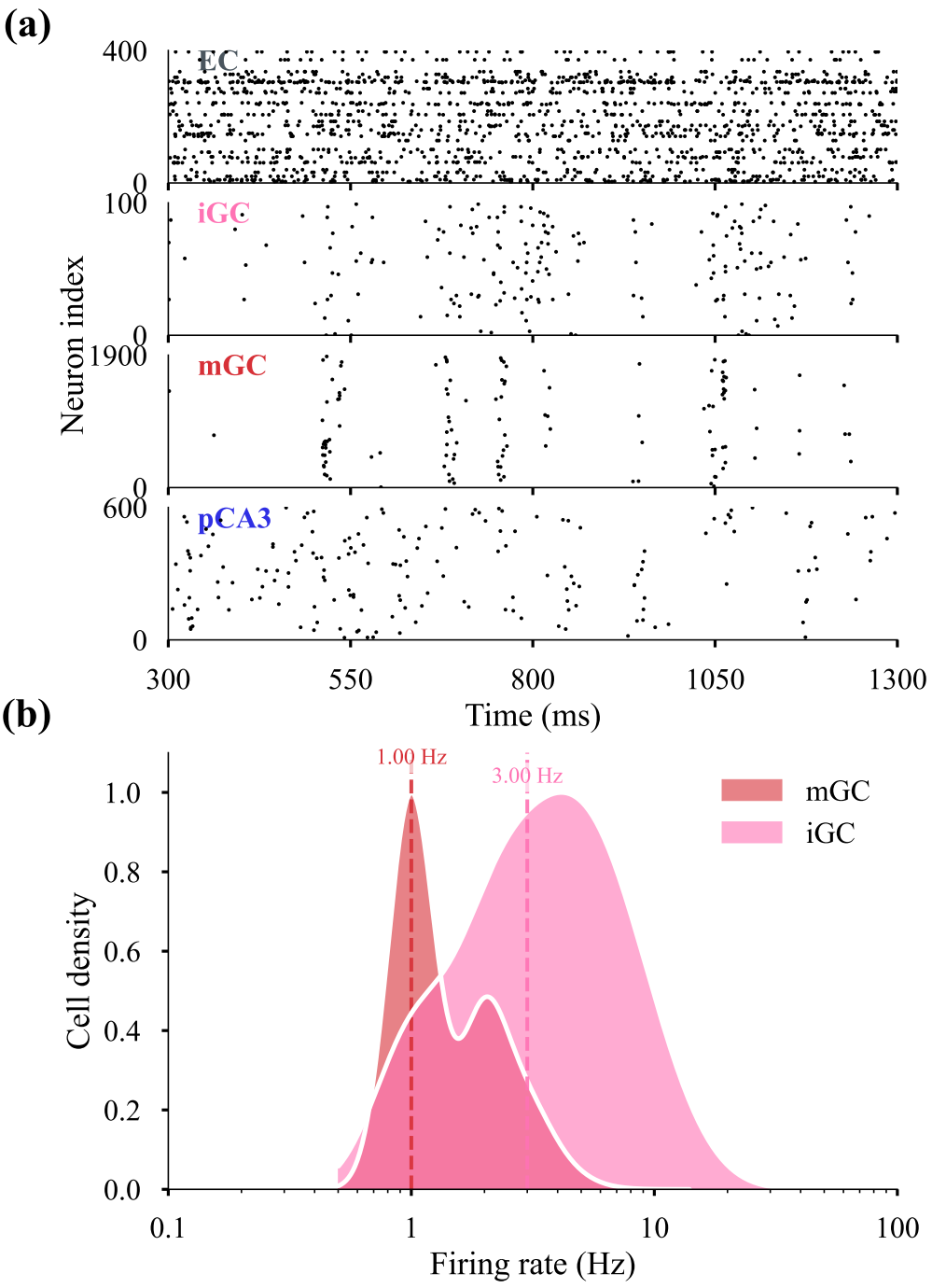
Before the effects of neurogenesis were analysed, it was first verified that the network settles into a stable and biologically plausible behavior. Figure 5a shows the spiking activity of the EC, iGC, mGC and pCA3 populations during a representative simulation. Driven by the EC input, the mGCs fire sparsely and in short, synchronized bouts, with only a small fraction of the population active at any moment. The iGCs are more active, while the pCA3 cells fire at a moderate rate, spread more evenly across the population.

This difference between the two populations can be seen in Figure 5b: the iGC firing-rate distribution is shifted towards higher rates than that of the mGCs, with a median firing rate of 3 Hz against 1 Hz for the mGCs. These values closely match those found *in vivo* by McHugh et al. (2022) (2.3 Hz for adult-born against 1.0 Hz for mGCs), although the shape of the distribution differs. For every cell type, the model's firing rates and population sparsity fell within the ranges compiled from the literature (Table 1), which is taken as evidence that the network behaves plausibly.

4.2 Adult neurogenesis enhances pattern separation in the DG and CA3

Figure 6a shows that, with neurogenesis, the mGC population activation drops, and with each increasing level of iGC excitability the mGC activation decreases further, from 7.9% in the control network to 3.6% at the highest excitability; the opposite is true for the iGC population, since higher levels of excitability are expected to make the iGCs more active, with almost 90% of the iGC population being active for the highest level of excitability simulated. This happens because of the feedforward inhibition exerted by iGCs upon mGCs, via iGC → mGC, iGC → BC/HIPP → mGC and iGC → MC → BC/HIPP → mGC. For increasing levels of iGC excitability, as mGC activity decreases and iGC activity increases, an equilibrium of approximately 8% of the entire GC population (iGCs and mGCs counted together) stays active, similarly to the control model without neurogenesis.

Figure 5 – Baseline network activity. **(a)** Raster plot of the spike times of the EC, iGC, mGC and pCA3 populations in a representative simulation, over the 300 ms to 1300 ms recording window (the first 300 ms discarded for stabilization). **(b)** Kernel-density estimate of the per-cell firing-rate distribution in logarithmic scale of the mGC and iGC populations; the dashed lines mark the population medians (mGC, 1 Hz; iGC, 3 Hz).



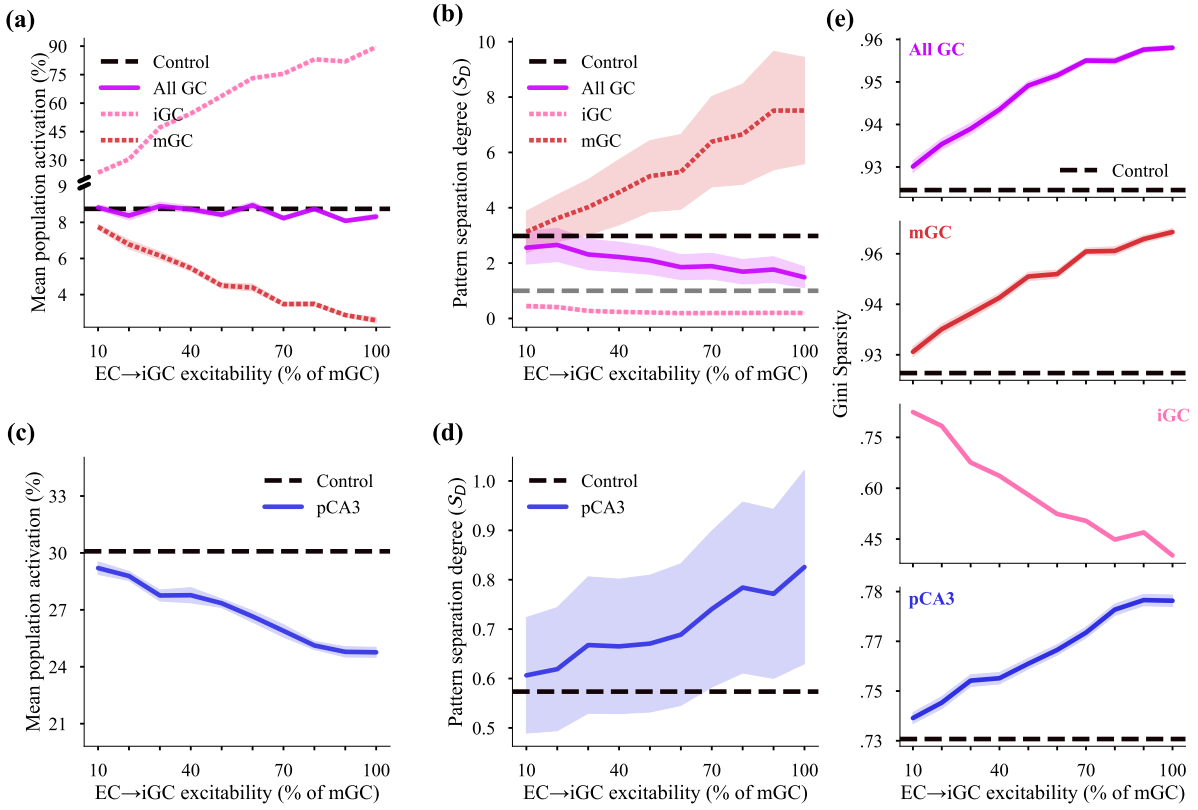
This lower activation seen in the mGC population with adult neurogenesis directly affects pattern separation: with fewer neurons active for a given pattern, the sparser coding leads to an increase in the mGC pattern separation degree, which rose monotonically with iGC excitability from $\mathcal{S}_D = 2.40$ in the control network to 5.80 at full excitability (Spearman $\rho = 1.00$, $p < 10^{-6}$; Figure 6b). Resolved by input similarity, the mGC separation showed the same ordering at every level, being strongest for the most similar inputs and higher than the control across the whole similarity range (Figure 8). The two GC types, however, behave in opposite ways. Taken on their own, the iGC outputs never separate their inputs ($\mathcal{S}_D < 1$, falling from 0.45 to 0.20 as excitability rises), acting instead as pattern integrators, as also reported by Kim and Lim (2024b). When the whole GC population is considered together, pattern separation remains above one but decreases with iGC excitability (from $\mathcal{S}_D = 2.40$ to 1.22; $\rho = -0.96$, $p < 10^{-5}$). Because the total GC activation stays near 8% across excitability levels, this decline is driven almost entirely by a loss of output decorrelation, the mean pairwise correlation between output patterns rising from 0.14 to 0.55, rather than by any change in activation. Since the role of iGCs is thought to be the modulation of network activity rather than the encoding of information (Fölsz et al., 2023; McHugh et al., 2022), the relevant quantity is the separation achieved by the encoding population, the mGCs, which is enhanced by neurogenesis.

The same analysis applied to the pCA3 population (Figures 6c and 6d) shows that its mean activation also decreases with neurogenesis and iGC excitability, from 29.7% to 24.1%, while its separation degree increases in the same direction as that of the mGCs, from $\mathcal{S}_D = 0.51$ to 0.69 ($\rho = +0.98$, $p < 10^{-5}$). Unlike the DG, however, the pCA3 separation degree remains below one throughout: by the definition described in Equation 3.12, the CA3 keeps its outputs more alike than its inputs. This is expected of a region whose recurrent connectivity supports autoassociation and pattern completion (Kopsick et al., 2024; Rolls, 2013) rather than separation. Neurogenesis therefore does not turn the CA3 into a separator, but pushes it toward separation.

4.3 Enhanced separation is driven mainly by increased sparsity

Figure 6e shows that the sparsity indices of both the DG and the CA3 increase with neurogenesis and with the progressive excitation of the iGCs, reproducing the central result

Figure 6 – Effects of adult neurogenesis and iGC excitability on activation, pattern separation and sparsity in the DG and CA3. In every panel the horizontal axis is the EC→iGC excitability level (10–100%), each point is the mean over the 30 pattern sets, shaded bands show the SEM and the dashed black line marks the control network without neurogenesis. **(a)** Mean population activation of DG GCs. **(b)** Pattern separation degree $\mathcal{S}_D$ of DG GCs; the grey dashed line marks $\mathcal{S}_D = 1$, above which the output patterns are more distinct than their inputs. **(c)** Mean population activation of the pCA3 cells. **(d)** Pattern separation degree of the pCA3 cells. **(e)** Population sparsity (Gini index) of the DG GCs and pCA3 cells.



of McHugh et al. (2022), where adult-born GCs promote population sparsity throughout the hippocampus. Across the ten excitability levels, the Gini index rose monotonically for the whole GC population (from 0.93 to 0.96; Spearman $\rho = +0.99$, $p < 10^{-6}$) and for the pCA3 cells (from 0.74 to 0.78; $\rho = +0.99$, $p < 10^{-6}$), each above its control value. The Gini index is shown here; the Hoyer measure, computed for cross-validation, follows the same trend (Figure 9).

The separation degree depends on both the orthogonalization degree and the activation degree of the patterns (Equations 3.8–3.12), so the mGCs could separate better by making their outputs sparser, more decorrelated, or both. Because the two combine multiplicatively, the change in $\log \mathcal{S}_D$ splits into an additive contribution from each. Most of the increase (about 92%) comes from the lower activation degree, as fewer mGCs fire for each pattern. The remaining

8% comes from a smaller but real gain in orthogonalization, as the mGC outputs become less correlated (Figure 10); this is not merely an effect of the sparser activity, since it remains when the number of active cells is held fixed. This extra decorrelation is likely due to the stronger winner-take-all competition driven by the iGCs (Coultrip et al., 1992). The entire GC population behaves the opposite: its decline in separation (Section 4.2) stems almost entirely (about 97%) from the increased correlation among its outputs, rather than from any change in activation.

4.4 iGC activation recruits circuit-wide inhibition

Directly injecting current into half of the iGCs propagated through the entire DG-CA3 network, as summarized by the PSTHs of Figure 7, pooled over all EC→iGC excitability variants.

Within the DG, the MCs were strongly excited and in turn drove strong excitation to the BCs. The BCs responded with a large, significant increase in firing, organized as a synchronous population rhythm at $\sim$ 127 Hz. The fast-spiking BCs are able to sustain a rhythm of this frequency, as their firing rate reaches up to $\sim$ 230 Hz *in vivo* (Lübke et al., 1998).

The HIPPs, by contrast, were only weakly inhibited (peak $z \approx -1$), a counterintuitive outcome given that they receive excitatory input from both the iGCs and the MCs; a plausible cause is that the transient drop in overall GC output following the pulse withdraws the GC→HIPP excitation that would otherwise engage them, outweighing the direct MC drive.

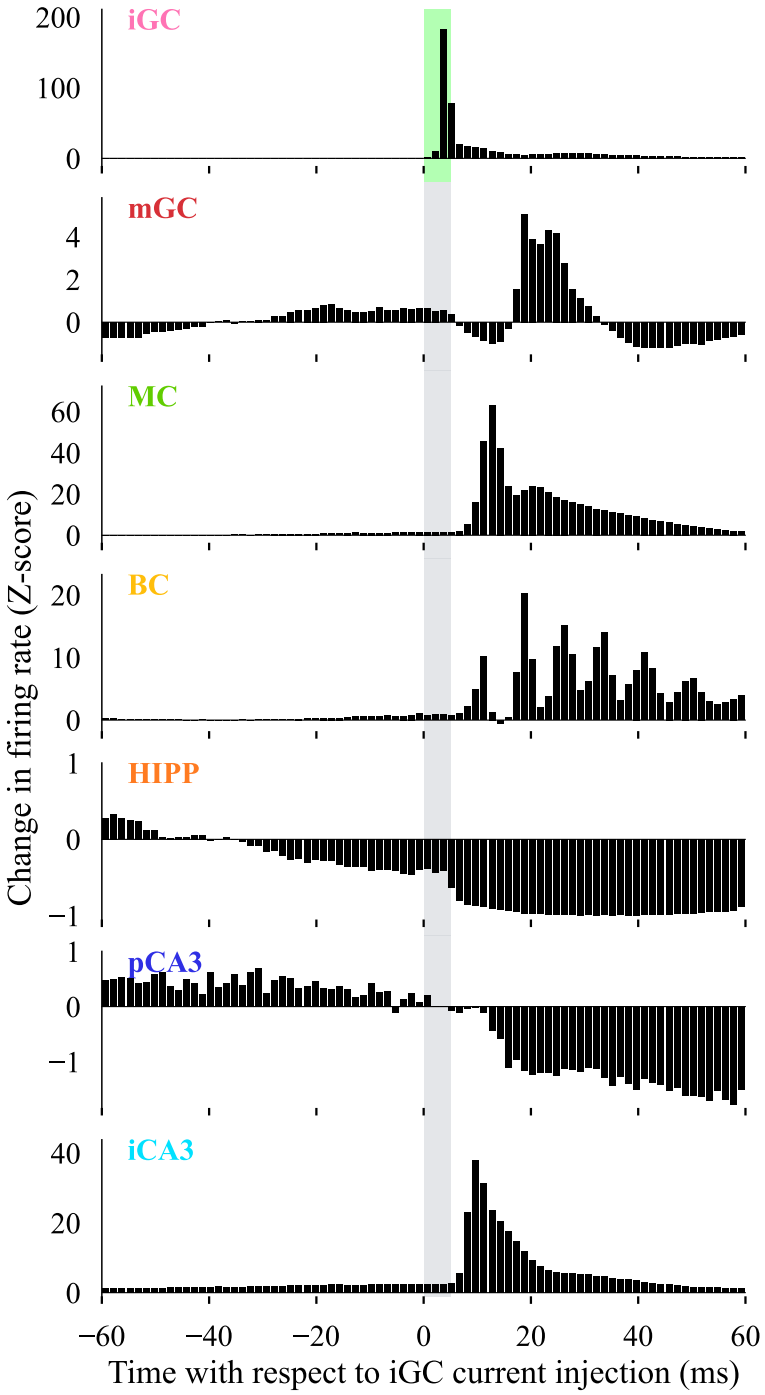
The mGCs displayed a biphasic response to the activation of the iGCs: a small excitation at first (peak $z \approx 4$), followed by an even smaller inhibition afterwards, with the net effect over the 60 ms being excitatory.

In CA3, the iCA3 interneurons were strongly and significantly activated, and this recruitment translated into a small inhibition of the pCA3 cells through the iCA3→pCA3 projection. McHugh et al. (2022) reported this interneuron recruitment to follow iGC spikes with a lag of 20 ms to 40 ms *in vivo*; because axonal conduction delays were not modeled and all synaptic delays were fixed at 1 ms, the absolute latencies here are shorter and not directly comparable, although the temporal ordering of the responses is preserved.

Comparing the Gini index of each population between the 50 ms windows immediately before and after the pulse confirmed that this recruitment of inhibition increased population

sparsity in most of the models, the within-trial counterpart of the steady-state sparsification reported in Section 4.3; the full comparison across populations and excitability levels is given in Figure 11. Overall, direct iGC activation recruited inhibitory interneurons across both the DG and CA3.

Figure 7 – Circuit-wide response to direct iGC activation. PSTHs of the population firing rate for each cell type, expressed as a z-score relative to a pre-pulse baseline (Equation 3.16) and aligned to the onset of the 5 ms depolarizing current injected into half of the iGCs. The shaded band marks the pulse, in green for the directly stimulated iGCs and in grey for every other population. Bins are 1.5 ms wide and span −60 ms to 60 ms relative to the pulse; data are pooled over all EC→iGC excitability variants and all trials. Vertical scales differ across panels. The directly driven iGCs (top) dominate during the pulse; downstream, the MCs, BCs and iCA3 cells show strong excitation, the pCA3 cells and HIPPs a weaker inhibition, and the mGCs a small biphasic response.



5 Discussion

Previous models of adult neurogenesis and pattern separation have been confined to the DG (Chavlis et al., 2017; Kim & Lim, 2024b; Yang et al., 2025), leaving open whether the circuit-wide sparsification observed *in vivo* (McHugh et al., 2022) follows from the known DG-CA3 microcircuitry. In the model built here, the DG reproduces the division of roles reported by Kim and Lim (2024b), the mGCs separating their inputs while the iGCs integrate them, with separation improving through sparser coding (Chavlis et al., 2017). The same computation, together with the interneuron recruitment it drives, extends to CA3, whose activity also becomes sparser, consistent with a modulatory rather than an encoding role for iGCs (Fölsz et al., 2023; McHugh et al., 2022). Independent evidence that the influence of neurogenesis is not confined to the DG comes from suppressing it: chemical or genetic ablation impairs the population-based coding of similar, but not dissimilar, contexts in CA3 (Niibori et al., 2012). Both features of that result are recovered here, in that the pCA3 separation degree rises with neurogenesis and the gain in mGC separation is largest for the most similar inputs (Figure 8), with the model supplying the circuit path by which a change confined to the DG reaches CA3.

The gain in mGC separation follows from a redistribution of activity within the GC population. Increasing the EC→iGC connectivity raises the iGC activation, and the iGCs drive the BCs and the HIPPs both directly and through the MCs; the resulting inhibition lowers the mGC activation, and the sparser mGC output raises its separation degree. Since the total GC activation stays close to the control value at every connectivity level, the DG does not become less active, and the activity given up by the mGCs is taken up by the iGCs. The separation degree measured over the whole GC population moves in the opposite direction for the same reason: the iGCs are few and are active for a large share of the patterns, so they are shared between patterns, and pooling them with the mGCs adds correlation to the output. That decline is accounted for almost entirely by the loss of decorrelation, the activation term showing no trend across connectivity levels. Kim and Lim (2024b) varied the same quantity and found two states divided by a threshold, the separation degree falling below that of a network without iGCs at low synaptic maturity, where the reduced innervation of the iGCs lowers their activation enough

to withdraw inhibition from the mGCs. No such state appears here, the mGC separation degree staying above the control at every level. The difference is consistent with how the two models reduce the input to the iGCs: Kim and Lim (2024b) scale the connection probability from the EC and the MCs together, whereas here only the EC→iGC probability is scaled and the MC→iGC projection is left intact, so the iGCs retain enough excitatory drive to engage the BCs and the HIPPs even at the lowest EC→iGC connectivity.

The pCA3 separation degree rises with neurogenesis but stays below one. Of the two factors combined in Equation 3.8, it is the activation term that holds it there: the pCA3 population is active at 24% to 30% against the 10% of the input patterns, and the measure returns values below one for any code denser than its input, whichever way the correlations move. Measured against the GC output that drives it rather than against the EC input, the pCA3 code is the more correlated of the two at low EC→iGC excitability and the less correlated at high ones, its own output correlation staying between 0.22 and 0.28 while that of the GC patterns rises from 0.14 to 0.55. The balance between the two factors also differs from the one found in the DG, the rise in the pCA3 separation degree being divided roughly two to one between the fall in activation and the gain in orthogonalization, against a much larger share for activation in the mGCs.

In agreement with McHugh et al. (2022), activating the iGCs recruited inhibitory interneurons in both the DG and CA3, the circuit counterpart of the increased GC population sparsity of Section 4.3. This recruitment was not a uniform elevation of firing: of all the populations, only the BCs show a spectral peak standing clearly above the background of their own spectrum (Figure 12), so the perisomatic inhibition engaged by the pulse is organized as a population rhythm. The analysis window resolves that frequency only to within one bin, so it should be taken as approximate.

How this relates to the one existing measurement of DG rhythmicity under altered neurogenesis is not straightforward. Eliminating the young adult-born population by focal irradiation or pharmacogenetic ablation reduced perforant-path evoked responses but *increased* the amplitude of spontaneous gamma bursts in the DG and hilus, and tightened the synchronization of dentate firing to them, which was taken as evidence that young neurons normally inhibit or destabilize recurrent activity in the region (Lacefield et al., 2012). Here, driving the iGCs

produced a synchronous basket cell rhythm rather than abolishing one. The two observations are not directly comparable: one is a chronic loss of function assessed on spontaneous activity, the other an acute gain of function assessed on an evoked response, and the frequency bands differ. They are at least consistent in implicating the iGCs in the control of perisomatic inhibition and of dentate synchrony, but the model cannot decide between them, because it contains no counterpart of the ablated condition: the control network here lacks iGCs from the outset rather than having them removed after they had integrated. Simulating that ablation, and measuring spontaneous rather than evoked rhythmicity, would make the comparison a real test.

Two features of the model response differ from the *in vivo* findings: the mGCs showed a net excitation rather than the suppression expected of a principal cell, and the pCA3 cells showed a small inhibition, for which McHugh et al. (2022) reported no effect. The area-level recordings of that study could not separate the inhibitory contributions of the HIPPs and the BCs, so the difference between the two DG interneuron populations cannot be tested against their data.

Neither the direct iGC→mGC inhibitory synapse nor the feedforward and feedback inhibition triggered by the pulse suppressed the mGCs. This residual excitation suggests that the iGC→mGC interaction is captured only in part, and that an inhibitory contribution is missing, plausibly from interneuron types absent from the reduced circuit: at least 13 morpho-physiological interneuron types have been distinguished in the DG (Degro et al., 2022), against the two represented here.

The size of the effect produced by the pulse depends on how much the iGCs already contribute at baseline. The same current is delivered to the same number of iGCs at every level and adds a roughly constant increment to the iGC population rate, while the baseline iGC rate rises about elevenfold across the connectivity range, so the pulse raises that rate some thirtyfold at the lowest connectivity and about fourfold at the highest. The gain in GC population sparsity produced by the pulse falls over the same range (Figure 11). The sparsification that follows a brief activation of adult-born GCs would therefore be expected to depend on how far the activated cells have integrated into the circuit, and to be largest for the least mature of them.

Several simplifications limit these results. Each neuron is reduced to a single compartment, so the dendritic processing that controls GC sparsity and separation (Chavlis et al., 2017), and

that varies across GC subtypes (Mao et al., 2025), is not represented. The interneuron diversity is minimal, with a single BC and HIPP type in the DG and one inhibitory type in CA3 (Kopsick et al., 2024), a likely source of the unexplained mGC response. The iGCs are represented as a single static population, so the sweep over EC→iGC connectivity samples the maturation range rather than following cells through it.

The effects on CA3 require further investigation. Because its synapses are held fixed here, the model cannot address how adult neurogenesis affects storage and recall in the recurrent CA3 network; introducing synaptic plasticity would allow this to be studied, along with the influence of adult-born cells on the formation and retrieval of CA3 engrams (Ko et al., 2025; Kopsick et al., 2024; Rolls, 2013).

6 Conclusion

This work set out to test whether the circuit-wide sparsification that follows the activation of adult-born granule cells *in vivo* (McHugh et al., 2022) can be accounted for by the DG-CA3 microcircuitry as it is currently characterized. The question was posed because the existing computational literature on neurogenesis and pattern separation stops at the DG (Chavlis et al., 2017; Kim & Lim, 2024b; Yang et al., 2025), and so could not address effects reported downstream, and because the candidate circuit mechanisms proposed for adult-born cells cannot be distinguished from one another in a model that excludes CA3 (Aimone, 2016).

To address it, a spiking model of the DG-CA3 network was assembled from the Hippocampome.org knowledge base (Wheeler et al., 2023) and extended with a population of iGCs whose parameters were fitted to published electrophysiological recordings (Espósito et al., 2005). Two experiments were then run on it. In the first, the level of neurogenesis and the excitability of the iGCs were varied while pattern separation and population sparsity were measured in the DG and in CA3. In the second, the iGCs were activated directly and the response of every population was recorded, as in the *in vivo* experiment that motivated the work.

Three findings follow. The mGC population became sparser and separated its inputs better as neurogenesis and iGC excitability increased, while the iGCs themselves never separated their own inputs and acted instead as integrators, so that the granule cell population considered as a whole moved in the opposite direction to its mature component. The pCA3 population followed the mGCs rather than the iGCs, becoming both sparser and more separating, although it continued to converge its inputs overall, as is expected of a region organized for autoassociation. Directly activating the iGCs recruited inhibitory interneurons in both the DG and CA3 and raised population sparsity, reproducing the effect observed *in vivo*. Decomposing the gain in mGC separation showed that most of it came from the reduction in the number of active cells rather than from a change in how correlated their outputs were, which identifies sparsification as the mechanism responsible rather than orthogonalization.

These results speak to the question of what adult-born granule cells are for. They are consistent with the view of the iGCs as transient modulators of the surrounding circuit rather

than as encoders in their own right (Fölsz et al., 2023; McHugh et al., 2022), and they attribute the effect to the circuitry rather than to plasticity, since the model reproduces it with synapses that never change. They also show that the effect is not confined to the region in which the new cells sit: a population of only 5% of the granule cells was enough to carry both the sparsification and the improvement in separation into CA3.

The conclusions are limited by the simplifications set out in Chapter 5. Neurons are reduced to single compartments, so the dendritic processing known to control GC sparsity is absent (Chavlis et al., 2017); the interneuron repertoire comprises two types in the DG and one in CA3, against the much larger diversity described anatomically (Degro et al., 2022); and the iGCs are represented as a single static population, so the sweep over their excitability samples the maturation range rather than following cells through it. The most important of these is the absence of synaptic plasticity, which is what prevents the model from addressing storage and recall in the recurrent CA3 network and therefore leaves the influence of adult-born cells on CA3 engrams open (Ko et al., 2025; Kopsick et al., 2024). Introducing plasticity, adding more interneuron types, and letting the iGC population mature over time are the natural next steps for this work.

A population of iGCs making up 5% of the GCs was enough to sparsen the mGC code, raise its pattern separation, and carry both effects into the CA3, while separating none of its own inputs. Regulation of coding sparsity across both regions therefore rests on the DG-CA3 microcircuitry alone, without synaptic plasticity, with the iGCs acting as modulators of activity rather than as encoders.

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APPENDIX A – Supplementary figures

This appendix collects the supporting analyses referred to in the Results and Discussion chapters. They are secondary to the main argument but are reported here so that the results can be checked in full: the pattern separation of the mGCs resolved by input similarity (Figure 8), the population sparsity measured with the Hoyer index as a cross-validation of the Gini results (Figure 9), the orthogonalization of the mGC output (Figure 10), the population sparsity immediately before and after direct iGC activation for each excitability level (Figure 11), and the post-pulse population rate and power spectrum of every cell type (Figure 12).

Figure 8 – Pattern separation of the mature granule cells resolved by input similarity. Each curve shows the pattern separation degree S_D of the mGC population as a function of the similarity between the input patterns, for the control network without neurogenesis (black dashed line) and for neurogenesis networks at representative EC→iGC excitability levels. Each point is the mean over the 30 pattern sets and the shaded band shows the standard error of the mean. The grey dashed line marks $S_D = 1$, above which the output patterns are more distinct than their inputs. Separation is strongest for the most similar (most overlapping) inputs, and every neurogenesis network separates better than the control across the whole similarity range.

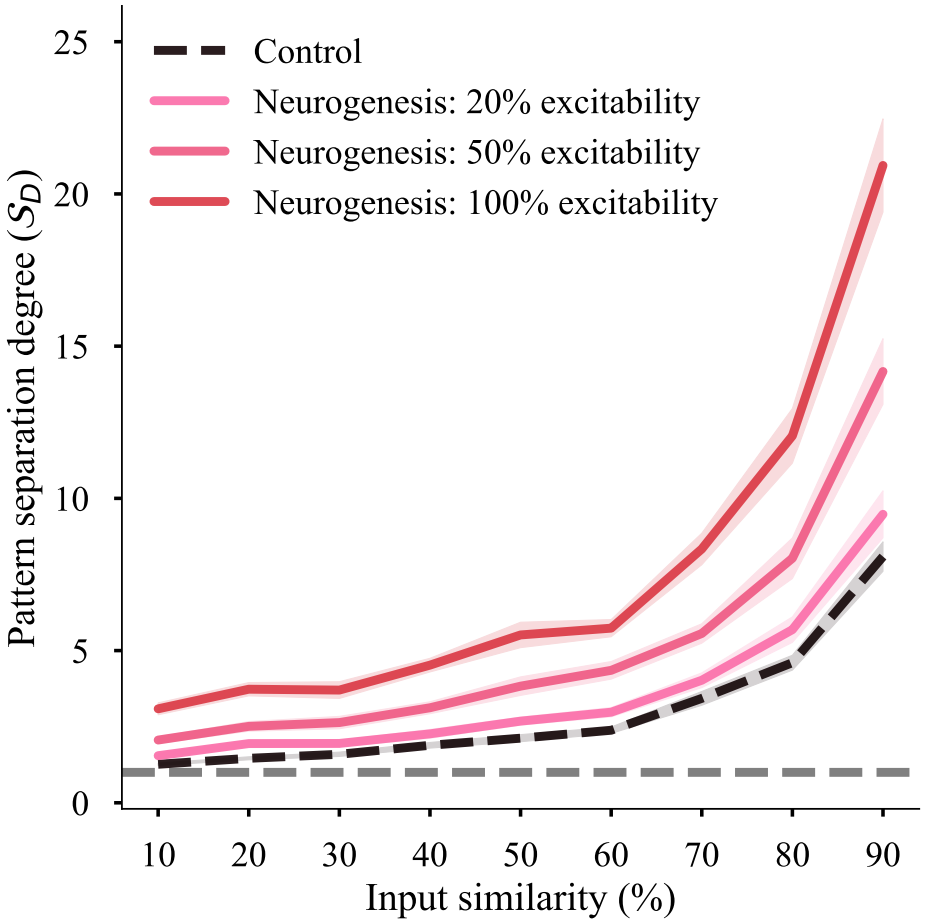


Figure 9 – Population sparsity measured with the Hoyer index, as a cross-validation of the Gini results reported in the Results chapter. Each panel shows the Hoyer sparsity of one population, the whole granule-cell population (All GC), the mature granule cells (mGC), the immature granule cells (iGC) and the pyramidal CA3 cells (pCA3), as a function of the EC→iGC excitability level; the shaded bands show the standard error of the mean and the dashed line marks the control network without neurogenesis (absent for the iGC panel, which has no counterpart in the control). As with the Gini index, the sparsity of the All GC, mGC and pCA3 populations rises with iGC excitability and stays above the control, whereas the iGC population becomes less sparse.

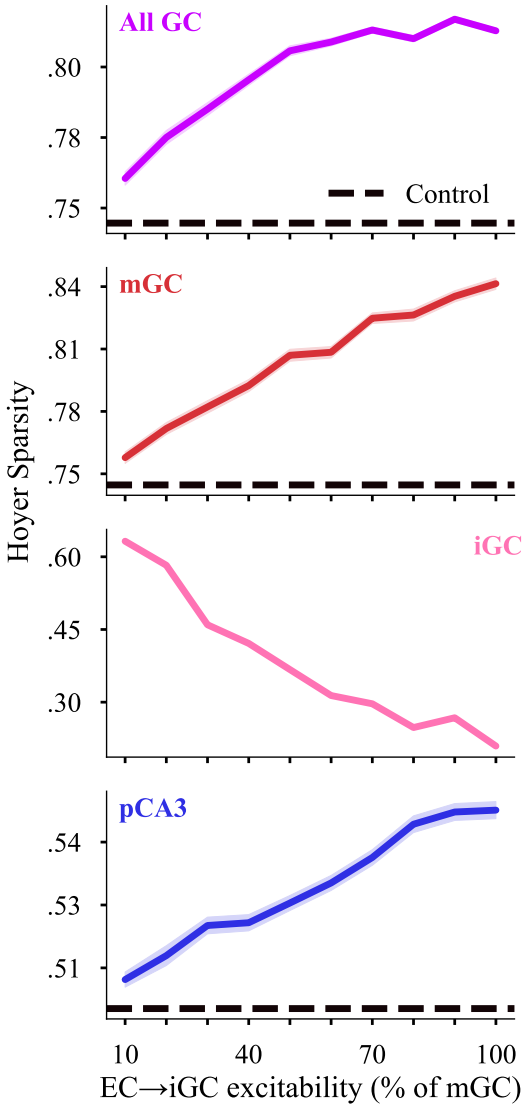


Figure 10 – Orthogonalization of the mature granule-cell output as a function of EC→iGC excitability. The curve shows the orthogonalization degree O_D of the mGC population, taken as the ratio of the output to the input orthogonalization ($O^{(out)}/O^{(in)}$); the shaded band shows the standard error of the mean and the dashed line marks the control network without neurogenesis. The orthogonalization rises with iGC excitability, indicating that the mGC outputs become increasingly decorrelated as neurogenesis increases, beyond what the reduction in the number of active cells alone would produce.

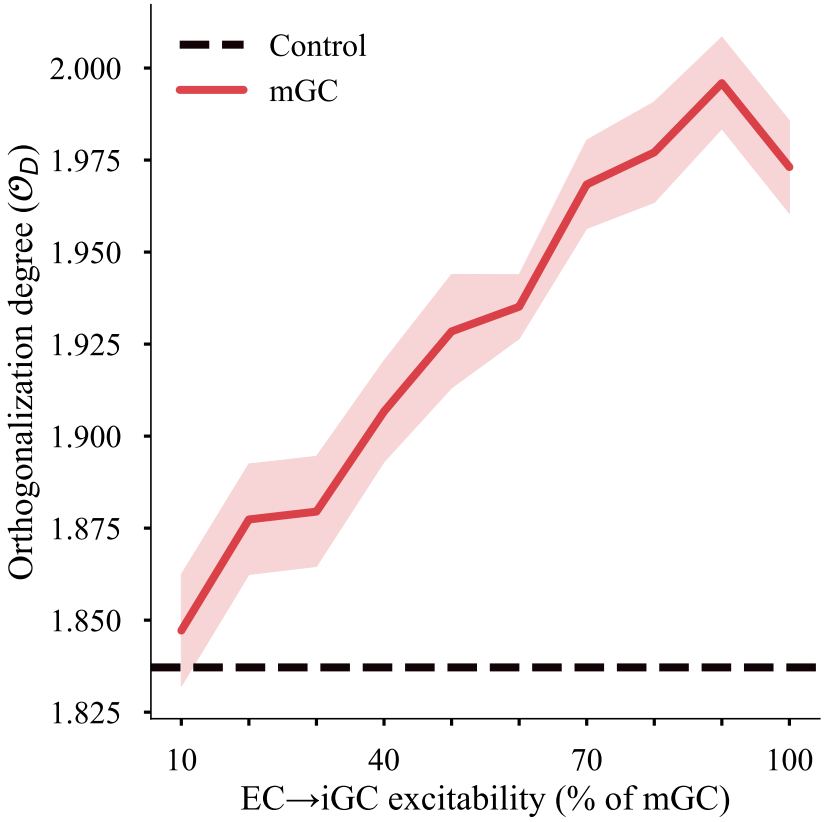


Figure 11 – Population sparsity (Gini index) immediately before and after direct iGC activation, for every EC→iGC excitability level. Each panel corresponds to one excitability level (10%–100%, given in the panel title) and compares the Gini index of the mGC, iGC, combined granule-cell (all GC) and pCA3 populations in the 50 ms windows immediately before (pre) and after (post) the depolarizing pulse, pooled over all runs. Bars show the mean across runs. The result of the paired Wilcoxon signed-rank test between the pre and post values is indicated above each pair (ns, not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$).

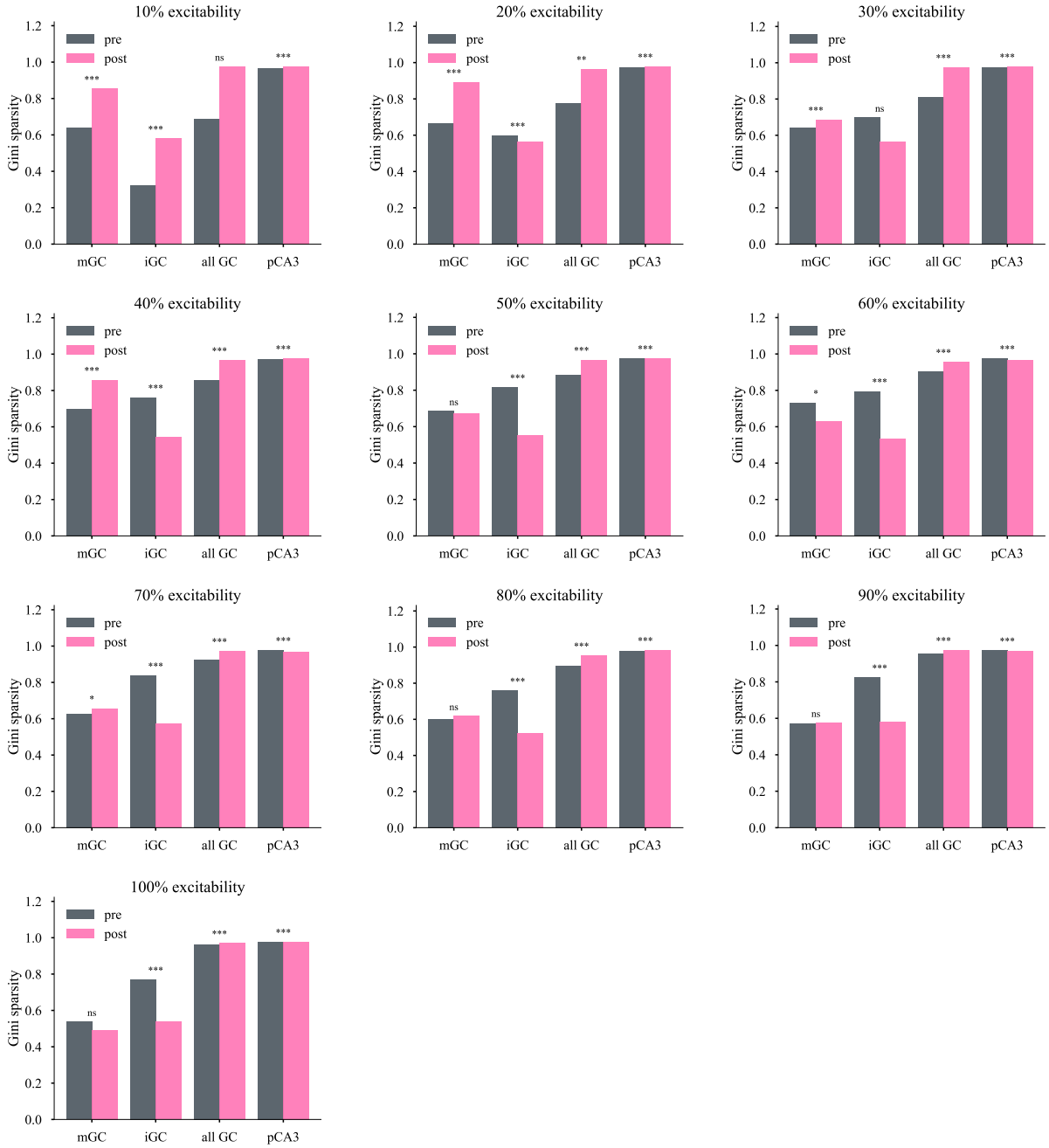


Figure 12 – Rhythmicity of the post-pulse response of each population. Left column, population firing rate over the 5 ms to 60 ms window following the onset of the depolarizing pulse delivered to half of the iGCs, pooled over all EC→iGC excitability levels and all runs. Right column, power spectrum of the same signal after subtraction of a Gaussian-smoothed envelope, estimated by Welch's method over the 20 Hz to 200 Hz band; the dashed line and the label mark the peak frequency. Only the BCs show a spectral peak that stands out from the background of their own spectrum, at the frequency reported in the Results chapter; the peak of the MCs carries a power close to the median of their band and the remaining populations peak at the lower edge of the band. The 55 ms analysis window sets the frequency resolution at 18 Hz, so the peak frequencies are accurate only to within one bin.

